

■ 시나리오 별 백업 복구(rman using recovery catalog)

KEY WORDS

- <시나리오 1> Setting Recovery catalog
- <시나리오 2> DF의 유실로 인한 복구
- <시나리오 3> 일반 TS를 다른 위치에 복구
- <시나리오 4> System DF 손상되었을 경우 복구
- <시나리오 5> 모든 DFs 손상되었을 경우 복구
- <시나리오 6> 모든 DFs의 손상으로 새로운 위치로 복구
- <시나리오 7> 백업받지 않은 TS의 DF 손상 복구
- <시나리오 8> TS에 속한 여러 DF 중 일부분이 손상된 경우 복구

<<시나리오 1>> Setting Recovery catalog

■ 네트워크 설정

<listener.ora>

```
[oracle@oel7v9r1 ~]$ vi $ORACLE_HOME/network/admin/listener.ora
```

```
SID_LIST_LISTENER =
```

```
(SID_LIST =
```

```
(SID_DESC =
```

```
(GLOBAL_DBNAME = orcl.localdomain)
```

```
(ORACLE_HOME = /u01/app/oracle/product/19.3.0/dbhome_1)
```

```
(SID_NAME = orcl)
```

```
)
```

```
(SID_DESC =
```

```
(GLOBAL_DBNAME = prod.localdomain)
```

```
(ORACLE_HOME = /u01/app/oracle/product/19.3.0/dbhome_1)
```

```
(SID_NAME = prod)
```

```
)
```

```
)
```

```
LISTENER =
```

```
(DESCRIPTION_LIST =
```

```
(DESCRIPTION =
```

```
(ADDRESS = (PROTOCOL = TCP)(HOST = oel7v9r1.localdomain)(PORT = 1521))
```

```
)
```

```
(DESCRIPTION =
```

```
(ADDRESS = (PROTOCOL = IPC)(KEY = EXTPROC1521))
```

```
)
```

```
)
```

```
ADR_BASE_LISTENER = /u01/app/oracle
```

<tnsnames.ora>

```
[oracle@oel7v9r1 ~]$ vi $ORACLE_HOME/network/admin/tnsnames.ora
```

```
T1 =
  (DESCRIPTION =
    (ADDRESS_LIST =
      (ADDRESS = (PROTOCOL = TCP)(HOST = oel7v9r1.localdomain)(PORT = 1521))
    )
    (CONNECT_DATA =
      (SERVICE_NAME = orcl.localdomain)
    )
  )
```

```
C1 =
  (DESCRIPTION =
    (ADDRESS_LIST =
      (ADDRESS = (PROTOCOL = TCP)(HOST = prod.localdomain)(PORT = 1521))
    )
    (CONNECT_DATA =
      (SERVICE_NAME = prod.localdomain)
    )
  )
```

```
LISTENER_ORCL =
  (ADDRESS = (PROTOCOL = TCP)(HOST = orcl.localdomain)(PORT = 1521))
```

■ Catalog DB에서

```
SYS@prod> create tablespace rts datafile '/u01/app/oracle/oradata/ORCL/rts01.dbf' size 300m;
Tablespace created.
```

```
SYS@prod> create user ruser identified by ruser
  2  default tablespace rts
  3  quota unlimited on rts;
User created.
```

```
SYS@prod> grant connect, resource, recovery_catalog_owner to ruser;
Grant succeeded.
```

```
SYS@prod> conn ruser/ruser
Connected.
```

■ Rman 접속

```
[oracle@oel7v9r1 admin]$ rman target "'sys/oracle@t1 as sysdba'" catalog ruser/ruser@c1
connected to target database: ORCL (DBID=1727714120)
connected to recovery catalog database
```

■ Recovery catalog 설정

```
RMAN> create catalog;
```

recovery catalog created

RMAN> **register database;**

database registered in recovery catalog

starting full resync of recovery catalog

full resync complete

RMAN> **report need backup;**

RMAN retention policy will be applied to the command

RMAN retention policy is set to redundancy 1

Report of files with less than 1 redundant backups

File #bkps Name

```
-----  
1    0    /u01/app/oracle/oradata/ORCL/system01.dbf  
3    0    /u01/app/oracle/oradata/ORCL/sysaux01.dbf  
4    0    /u01/app/oracle/oradata/ORCL/undotbs01.dbf  
7    0    /u01/app/oracle/oradata/ORCL/users01.dbf
```

RMAN> **host 'mkdir -p /home/oracle/backup/FRA';**

host command complete

RMAN> **sql 'alter system set db_recovery_file_dest="/home/oracle/backup/FRA" scope=spfile';**

sql statement: alter system set db_recovery_file_dest="/home/oracle/backup/FRA" scope=spfile

RMAN> **sql 'alter system set db_recovery_file_dest_size=4G scope=spfile';**

sql statement: alter system set db_recovery_file_dest_size=4G scope=spfile

RMAN> shut immediate

database closed

database dismounted

Oracle instance shut down

RMAN> startup

connected to target database (not started)

Oracle instance started

database mounted

database opened

Total System Global Area 1140849904 bytes

Fixed Size 8895728 bytes

Variable Size 822083584 bytes

Database Buffers 301989888 bytes
 Redo Buffers 7880704 bytes

RMAN> **list incarnation;**

List of Database Incarnations

DB Key	Inc Key	DB Name	DB ID	STATUS	Reset SCN	Reset Time
1	16	ORCL	1727714120	PARENT	1	17-APR-19
1	17	ORCL	1727714120	PARENT	1920977	01-MAY-25
1	18	ORCL	1727714120	PARENT	2310250	07-MAY-25
1	19	ORCL	1727714120	PARENT	2897913	13-MAY-25
1	2	ORCL	1727714120	CURRENT	3114462	15-MAY-25

RMAN> **select * from v\$database_incarnation;**

INCARNATION# RESETLOGS_CHANGE# RESETLOGS PRIOR_RESETLOGS_CHANGE# PRIOR_RES

STATUS	RESETLOGS_ID	PRIOR_INCARNATION#	FLASHBACK_DATABASE_ALLOWED	CON_ID
	1	1	17-APR-19	0
PARENT	1005785759	0	NO	0
	2	1920977	01-MAY-25	1
PARENT	1199933387	1	NO	0
	3	2310250	07-MAY-25	1920977
PARENT	1200525254	2	NO	0
	4	2897913	13-MAY-25	2310250
PARENT	1201040433	3	NO	0
	5	3114462	15-MAY-25	2897913
CURRENT	1201218512	4	NO	0

■ Archive log 확인

RMAN> **list archivelog all;**

List of Archived Log Copies for database with db_unique_name ORCL

Key	Thrd	Seq	S	Low Time	
199	1	12	A	15-MAY-25	Name: /home/oracle/arch2/arch_1_12_1201040433.arc
198	1	12	A	15-MAY-25	Name: /home/oracle/arch1/arch_1_12_1201040433.arc
195	1	13	A	15-MAY-25	Name: /home/oracle/arch2/arch_1_13_1201040433.arc
194	1	13	A	15-MAY-25	Name: /home/oracle/arch1/arch_1_13_1201040433.arc
197	1	14	A	15-MAY-25	Name: /home/oracle/arch2/arch_1_14_1201040433.arc
196	1	14	A	15-MAY-25	Name: /home/oracle/arch1/arch_1_14_1201040433.arc

ID	Channel	Device	Media	File Name	File Path
201	1	1	A	15-MAY-25	Name: /home/oracle/arch2/arch_1_1_1201218512.arc
200	1	1	A	15-MAY-25	Name: /home/oracle/arch1/arch_1_1_1201218512.arc
203	1	2	A	17-MAY-25	Name: /home/oracle/arch2/arch_1_2_1201218512.arc
202	1	2	A	17-MAY-25	Name: /home/oracle/arch1/arch_1_2_1201218512.arc
205	1	3	A	17-MAY-25	Name: /home/oracle/arch2/arch_1_3_1201218512.arc
204	1	3	A	17-MAY-25	Name: /home/oracle/arch1/arch_1_3_1201218512.arc
207	1	4	A	17-MAY-25	Name: /home/oracle/arch2/arch_1_4_1201218512.arc
206	1	4	A	17-MAY-25	Name: /home/oracle/arch1/arch_1_4_1201218512.arc

RMAN> **alter system archive log current;**

<추가>

ID	Channel	Device	Media	File Name	File Path
309	1	5	A	17-MAY-25	Name: /home/oracle/arch2/arch_1_5_1201218512.arc
308	1	5	A	17-MAY-25	Name: /home/oracle/arch1/arch_1_5_1201218512.arc

■ Retention policy 예

RMAN> **report obsolete;**

RMAN retention policy will be applied to the command

RMAN retention policy is set to redundancy 1

Report of obsolete backups and copies

Type	Key	Completion Time	Filename/Handle
Backup Set	208	15-MAY-25	
Backup Piece	209	15-MAY-25	/u01/app/oracle/product/19.3.0/dbhome_1/dbs/c-1727714120-20250515-02

RMAN> **delete obsolete;**

RMAN retention policy will be applied to the command

RMAN retention policy is set to redundancy 1

allocated channel: ORA_DISK_1

channel ORA_DISK_1: SID=84 device type=DISK

Deleting the following obsolete backups and copies:

Type	Key	Completion Time	Filename/Handle
Backup Set	208	15-MAY-25	
Backup Piece	209	15-MAY-25	/u01/app/oracle/product/19.3.0/dbhome_1/dbs/c-1727714120-20250515-02

Do you really want to delete the above objects (enter YES or NO)? **YES**

deleted backup piece

backup piece handle=/u01/app/oracle/product/19.3.0/dbhome_1/dbs/c-1727714120-20250515-02 RECID=12
STAMP=1201218516

Deleted 1 objects

RMAN> **report obsolete;**

RMAN retention policy will be applied to the command

RMAN retention policy is set to redundancy 1

no obsolete backups found

■ Catalog DB에서

RUSER@prod> **select * from rc_backup_set;**

RUSER@prod> **select * from rc_backup_piece;**

RUSER@prod> **select * from rc_backup_controlfile;**

RUSER@prod> **select * from RC_STORED_SCRIPT;**

no rows selected

RUSER@prod> **select * from rc_stored_script_line;**

no rows selected

RUSER@prod> **select db_name, dbinc_key, tablespace_name, name from rc_datafile;**

DB_NAME	DBINC_KEY	TABLESPACE_NAME	NAME
ORCL	2	SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf
ORCL	2	SYSAUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf
ORCL	2	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf
ORCL	2	USERS	/u01/app/oracle/oradata/ORCL/users01.dbf

ORCL	2	SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf
ORCL	2	SYSAUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf
ORCL	2	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf
ORCL	2	USERS	/u01/app/oracle/oradata/ORCL/users01.dbf

- rman에서

RMAN> **create tablespace hrtbs datafile '/u01/app/oracle/oradata/ORCL/hrtbs01.dbf' size 10m;**

Statement processed

starting full resync of recovery catalog

full resync complete

RUSER@prod> **select db_name, dbinc_key, tablespace_name, name from rc_datafile where dbinc_key = 2;**

ORCL	2	HRTBS	/u01/app/oracle/oradata/ORCL/hrtbs01.dbf	<--- 추가
------	---	-------	--	---------

rman에서

RMAN> **drop tablespace hrtbs including contents and datafiles;**

Statement processed

starting full resync of recovery catalog
full resync complete

```
RUSER@prod> select db_name, dbinc_key, tablespace_name, name from rc_datafile where dbinc_key = 2;
ORCL          2 HRTBS          /u01/app/oracle/oradata/ORCL/hrtbs01.dbf      <--- 남아 있음
```

rman에서

```
RMAN> resync catalog;
```

starting full resync of recovery catalog
full resync complete

```
RMAN> report schema;
```

Report of database schema for database with db_unique_name ORCL

List of Permanent Datafiles

=====

File	Size(MB)	Tablespace	RB segs	Datafile Name
1	910	SYSTEM	YES	/u01/app/oracle/oradata/ORCL/system01.dbf
3	590	SYSAUX	NO	/u01/app/oracle/oradata/ORCL/sysaux01.dbf
4	345	UNDOTBS1	YES	/u01/app/oracle/oradata/ORCL/undotbs01.dbf
7	5	USERS	NO	/u01/app/oracle/oradata/ORCL/users01.dbf

File	Size(MB)	Tablespace	Maxsize(MB)	Tempfile Name
1	32	TEMP	32767	/u01/app/oracle/oradata/ORCL/temp01.dbf

■ rman에서 (Script 사용)

```
[oracle@oel7v9r1 admin]$ rman catalog ruser/ruser@c1
connected to recovery catalog database
```

```
RMAN> list backup;
```

```
RMAN-00571: =====
RMAN-00569: ===== ERROR MESSAGE STACK FOLLOWS =====
RMAN-00571: =====
RMAN-03002: failure of list command at 05/17/2025 23:07:46
RMAN-06171: not connected to target database
RMAN> exit
```

```
[oracle@oel7v9r1 admin]$ rman target "'sys/oracle@t1 as sysdba'" catalog ruser/ruser@c1
```

```

RMAN> create script incr_0_backup{
2> backup as backupset incremental level 0 database include current controlfile;
3> sql 'alter system archive log current';}
created script incr_0_backup

```

```
RMAN> list script names;
```

List of Stored Scripts in Recovery Catalog

Scripts of Target Database ORCL

Script Name

Description

incr_0_backup

```
RMAN> print script incr_0_backup;
```

printing stored script: incr_0_backup

```

{
backup as backupset incremental level 0 database include current controlfile;
sql 'alter system archive log current';}

```

- recovery catalog db에서

```
RUSER@prod> select * from RC_STORED_SCRIPT;
```

DB_KEY	DB_NAME	SCRIPT_NAME	SCRIPT_COMMENT

1	ORCL	incr_0_backup	

```
RUSER@prod > select * from rc_stored_script_line;
```

DB_KEY	SCRIPT_NAME	LINE	TEXT

1	incr_0_backup	1	{ 1 incr_0_backup
		2	backup as backupset incremental level 0 database include current controlfile;
1	incr_0_backup	3	sql 'alter system archive log current';}

```
RMAN> replace script incr_0_backup{
```

```
2> backup as compressed backupset incremental level 0 database include current controlfile;
```

```
3> alter system archive log current;}
```

replaced script incr_0_backup

```
RMAN> run {execute script incr_0_backup;}
```

executing script: incr_0_backup

Starting backup at 17-MAY-25
 allocated channel: ORA_DISK_1
 channel ORA_DISK_1: SID=80 device type=DISK
 channel ORA_DISK_1: starting compressed incremental level 0 datafile backup set
 channel ORA_DISK_1: specifying datafile(s) in backup set
 input datafile file number=00001 name=/u01/app/oracle/oradata/ORCL/system01.dbf
 input datafile file number=00003 name=/u01/app/oracle/oradata/ORCL/sysaux01.dbf
 input datafile file number=00004 name=/u01/app/oracle/oradata/ORCL/undotbs01.dbf
 input datafile file number=00007 name=/u01/app/oracle/oradata/ORCL/users01.dbf
 channel ORA_DISK_1: starting piece 1 at 17-MAY-25
 channel ORA_DISK_1: finished piece 1 at 17-MAY-25
 piece
 handle=/home/oracle/backup/FRA/ORCL/backupset/2025_05_17/o1_mf_nnnd0_TAG20250517T232101_n2k6tffz_bk
 p tag=TAG20250517T232101 comment=NONE
 channel ORA_DISK_1: backup set complete, elapsed time: 00:00:35
 channel ORA_DISK_1: starting compressed incremental level 0 datafile backup set
 channel ORA_DISK_1: specifying datafile(s) in backup set
 including current control file in backup set
 channel ORA_DISK_1: starting piece 1 at 17-MAY-25
 channel ORA_DISK_1: finished piece 1 at 17-MAY-25
 piece
 handle=/home/oracle/backup/FRA/ORCL/backupset/2025_05_17/o1_mf_ncnn0_TAG20250517T232101_n2k6vkkt_b
 kp tag=TAG20250517T232101 comment=NONE
 channel ORA_DISK_1: backup set complete, elapsed time: 00:00:01
 Finished backup at 17-MAY-25

Starting Control File and SPFILE Autobackup at 17-MAY-25
 piece handle=/home/oracle/backup/FRA/ORCL/autobackup/2025_05_17/o1_mf_s_1201389699_n2k6vmpx_.bkp
 comment=NONE
 Finished Control File and SPFILE Autobackup at 17-MAY-25

Statement processed

RMAN> **list backup;**

List of Backup Sets

=====

BS Key	Type	LV	Size	Device	Type	Elapsed Time	Completion Time
215	Full		10.30M	DISK		00:00:00	17-MAY-25
BP Key: 218 Status: AVAILABLE Compressed: NO Tag: TAG20250517T213923							
Piece Name: /u01/app/oracle/product/19.3.0/dbhome_1/dbs/c-1727714120-20250517-00							

SPFILE Included: Modification time: 17-MAY-25

SPFILE db_unique_name: ORCL

Control File Included: Ckp SCN: 3333307 Ckp time: 17-MAY-25

BS Key	Type	LV Size	Device	Type	Elapsed Time	Completion Time
--------	------	---------	--------	------	--------------	-----------------

368	Full	10.30M	DISK		00:00:00	17-MAY-25
-----	------	--------	------	--	----------	-----------

BP Key: 371 Status: AVAILABLE Compressed: NO Tag: TAG20250517T230516

Piece

Name:

/home/oracle/backup/FRA/ORCL/autobackup/2025_05_17/o1_mf_s_1201388716_n2k5ww7h_.bkp

SPFILE Included: Modification time: 17-MAY-25

SPFILE db_unique_name: ORCL

Control File Included: Ckp SCN: 3338842 Ckp time: 17-MAY-25

BS Key	Type	LV Size	Device	Type	Elapsed Time	Completion Time
--------	------	---------	--------	------	--------------	-----------------

388	Incr 0	294.94M	DISK		00:00:29	17-MAY-25
-----	--------	---------	------	--	----------	-----------

BP Key: 392 Status: AVAILABLE Compressed: YES Tag: TAG20250517T232101

Piece

Name:

/home/oracle/backup/FRA/ORCL/backupset/2025_05_17/o1_mf_nnnd0_TAG20250517T232101_n2k6tffz_.bkp

List of Datafiles in backup set 388

File	LV	Type	Ckp SCN	Ckp Time	Abs Fuz SCN	Sparse	Name
------	----	------	---------	----------	-------------	--------	------

1	0	Incr	3340118	17-MAY-25		NO	/u01/app/oracle/oradata/ORCL/system01.dbf
3	0	Incr	3340118	17-MAY-25		NO	/u01/app/oracle/oradata/ORCL/sysaux01.dbf
4	0	Incr	3340118	17-MAY-25		NO	/u01/app/oracle/oradata/ORCL/undotbs01.dbf
7	0	Incr	3340118	17-MAY-25		NO	/u01/app/oracle/oradata/ORCL/users01.dbf

BS Key	Type	LV Size	Device	Type	Elapsed Time	Completion Time
--------	------	---------	--------	------	--------------	-----------------

389	Incr 0	1.05M	DISK		00:00:01	17-MAY-25
-----	--------	-------	------	--	----------	-----------

BP Key: 393 Status: AVAILABLE Compressed: YES Tag: TAG20250517T232101

Piece

Name:

/home/oracle/backup/FRA/ORCL/backupset/2025_05_17/o1_mf_ncnn0_TAG20250517T232101_n2k6vkkt_.bkp

Control File Included: Ckp SCN: 3340133 Ckp time: 17-MAY-25

BS Key	Type	LV Size	Device	Type	Elapsed Time	Completion Time
--------	------	---------	--------	------	--------------	-----------------

403	Full	10.30M	DISK		00:00:00	17-MAY-25
-----	------	--------	------	--	----------	-----------

BP Key: 405 Status: AVAILABLE Compressed: NO Tag: TAG20250517T232139

Piece

Name:

/home/oracle/backup/FRA/ORCL/autobackup/2025_05_17/o1_mf_s_1201389699_n2k6vmpx_.bkp

SPFILE Included: Modification time: 17-MAY-25

SPFILE db_unique_name: ORCL

Control File Included: Ckp SCN: 3340148 Ckp time: 17-MAY-25

- Target db에서

SYS@orcl> **select * from v\$flash_recovery_area_usage;**

SYS@orcl> **!ls -al /home/oracle/backup/FRA/ORCL/backupset/2025_05_17**

SYS@orcl> **!ls -al /home/oracle/backup/FRA/ORCL/autobackup/2025_05_17**

SYS@orcl> **select * from v\$recovery_file_dest;**

NAME

SPACE_LIMIT	SPACE_USED	SPACE_RECLAIMABLE	NUMBER_OF_FILES	CON_ID

/home/oracle/backup/FRA				
4294967296	331956224	11894784	4	0

SYS@orcl> alter system reset log_archive_dest_2 scope=spfile;

<--- 초기화

System altered.

RMAN> **delete script incr_0_backup;**

deleted script: incr_0_backup

RMAN> **list script names;**

List of Stored Scripts in Recovery Catalog

No scripts in recovery catalog

- Recovery catalog db에서

RUSER@prod> **select * from rc_stored_script;**

no rows selected

<<시나리오 2>> DF 의 유실로 인한 복구

```
[oracle@oel7v9r1 admin]$ rman target "'sys/oracle@t1 as sysdba'" catalog ruser/ruser@c1
```

```
RMAN> select dbid, name, log_mode from v$database;
```

DBID	NAME	LOG_MODE
1727714120	ORCL	ARCHIVELOG

```
RMAN> select file_id, tablespace_name, file_name, bytes/1024/1024 mb from dba_data_files;
```

FILE_ID	TABLESPACE_NAME	FILE_NAME	MB
1	SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf	910
3	SYSAUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	590
7	USERS	/u01/app/oracle/oradata/ORCL/users01.dbf	5
4	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	345

```
RMAN> select file_id, tablespace_name, file_name, bytes/1024/1024 mb from dba_temp_files;
```

FILE_ID	TABLESPACE_NAME	FILE_NAME	MB
1	TEMP	/u01/app/oracle/oradata/ORCL/temp01.dbf	32

```
RMAN> report schema;
```

```
RMAN> show all;
```

RMAN configuration parameters for database with db_unique_name ORCL are:

CONFIGURE RETENTION POLICY TO REDUNDANCY 1; # default

CONFIGURE BACKUP OPTIMIZATION OFF; # default

CONFIGURE DEFAULT DEVICE TYPE TO DISK; # default

CONFIGURE CONTROLFILE AUTOBACKUP ON; # default

CONFIGURE CONTROLFILE AUTOBACKUP FORMAT FOR DEVICE TYPE DISK TO '%F'; # default

CONFIGURE DEVICE TYPE DISK PARALLELISM 1 BACKUP TYPE TO BACKUPSET; # default

CONFIGURE DATAFILE BACKUP COPIES FOR DEVICE TYPE DISK TO 1; # default

CONFIGURE ARCHIVELOG BACKUP COPIES FOR DEVICE TYPE DISK TO 1; # default

CONFIGURE MAXSETSIZE TO UNLIMITED; # default

CONFIGURE ENCRYPTION FOR DATABASE OFF; # default

CONFIGURE ENCRYPTION ALGORITHM 'AES128'; # default

CONFIGURE COMPRESSION ALGORITHM 'BASIC' AS OF RELEASE 'DEFAULT' OPTIMIZE FOR LOAD TRUE ; # default

CONFIGURE RMAN OUTPUT TO KEEP FOR 7 DAYS; # default

CONFIGURE ARCHIVELOG DELETION POLICY TO NONE; # default

CONFIGURE SNAPSHOT CONTROLFILE NAME TO '/u01/app/oracle/product/19.3.0/dbhome_1/dbs/snapcf_orcl.f'; # default

```

RMAN> list backup;
RMAN> delete backup;
RMAN> list backup;

```

- Target db에서

```

[oracle@oel7v9r1 ORCL]$ pwd
/home/oracle/backup/FRA/ORCL
[oracle@oel7v9r1 ORCL]$ ls autobackup/2025_05_17/ backupset/2025_05_17/
autobackup/2025_05_17/:

```

```

backupset/2025_05_17/:

```

■ 백업

```

RMAN> report need backup;

```

```

RMAN> backup database;

```

```

RMAN> list backup;

```

List of Backup Sets

=====

BS Key	Type	LV	Size	Device	Type	Elapsed Time	Completion Time
--------	------	----	------	--------	------	--------------	-----------------

516	Full		1.21G	DISK		00:00:03	18-MAY-25
-----	------	--	-------	------	--	----------	-----------

BP Key: 517 Status: AVAILABLE Compressed: NO Tag: TAG20250518T001132

Piece

Name:

/home/oracle/backup/FRA/ORCL/backupset/2025_05_18/o1_mf_nnndf_TAG20250518T001132_n2k9s4d8_.bkp

List of Datafiles in backup set 516

File	LV	Type	Ckp SCN	Ckp Time	Abs Fuz SCN	Sparse	Name
------	----	------	---------	----------	-------------	--------	------

1		Full	3342795	18-MAY-25		NO	/u01/app/oracle/oradata/ORCL/system01.dbf
3		Full	3342795	18-MAY-25		NO	/u01/app/oracle/oradata/ORCL/sysaux01.dbf
4		Full	3342795	18-MAY-25		NO	/u01/app/oracle/oradata/ORCL/undotbs01.dbf
7		Full	3342795	18-MAY-25		NO	/u01/app/oracle/oradata/ORCL/users01.dbf

BS Key	Type	LV	Size	Device	Type	Elapsed Time	Completion Time
--------	------	----	------	--------	------	--------------	-----------------

532	Full		10.30M	DISK		00:00:00	18-MAY-25
-----	------	--	--------	------	--	----------	-----------

BP Key: 537 Status: AVAILABLE Compressed: NO Tag: TAG20250518T001140

Piece

Name:

/home/oracle/backup/FRA/ORCL/autobackup/2025_05_18/o1_mf_s_1201392700_n2k9sdbw_.bkp

SPFILE Included: Modification time: 17-MAY-25

SPFILE db_unique_name: ORCL

Control File Included: Ckp SCN: 3342812 Ckp time: 18-MAY-25

■ 복구 시나리오

- Target DB에서

```
SYS@orcl> create table hr.loc1 tablespace users as select * from hr.locations;
```

Table created.

```
SYS@orcl> alter system switch logfile; <--- 여러번 수행
```

System altered.

```
SYS@orcl> col name for a50
```

```
SYS@orcl> col member for a50
```

```
SYS@orcl> set linesize 200
```

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

GROUP#	SEQUENCE#	MEMBER	MB	STATUS
ARC	FIRST_CHANGE#	NEXT_CHANGE#		
1	10	/u01/app/oracle/oradata/ORCL/redo01.log	10	INACTIVE
3344458	3344461			YES
2	11	/u01/app/oracle/oradata/ORCL/redo02.log	10	CURRENT
3344461	1.8447E+19			NO
3	9	/u01/app/oracle/oradata/ORCL/redo03.log	10	INACTIVE
3344455	3344458			YES

```
SYS@orcl> select sequence#, name, FIRST_CHANGE#, NEXT_CHANGE# from v$archived_log;
```

```
SYS@orcl> select f.tablespace_name, f.file_name from dba_extents e, dba_data_files f where e.file_id = f.file_id and
e.segment_name='LOC1' and e.owner='HR';
```

```
TABLESPACE_NAME FILE_NAME
```

```
USERS /u01/app/oracle/oradata/ORCL/users01.dbf
```

- 장애 발생

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/users01.dbf
```

```
SYS@orcl> !ls /u01/app/oracle/oradata/ORCL/users01.dbf
```

ls: cannot access /u01/app/oracle/oradata/ORCL/users01.dbf: No such file or directory

SYS@orcl> select count(*) from hr.loc1;

COUNT(*)

23 <--- 메모리 상주, 질의 가능

SYS@orcl> alter system checkpoint;

System altered.

- Alert log file 확인

025-05-18T00:30:58.192260+09:00

Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl_mz01_21638.trc:

ORA-01110: data file 7: '/u01/app/oracle/oradata/ORCL/users01.dbf'

ORA-01565: error in identifying file '/u01/app/oracle/oradata/ORCL/users01.dbf'

ORA-27037: unable to obtain file status

Linux-x86_64 Error: 2: No such file or directory

Additional information: 7

RMAN> **list failure;**

Database Role: PRIMARY

List of Database Failures

=====

Failure ID	Priority	Status	Time Detected	Summary
------------	----------	--------	---------------	---------

2	HIGH	OPEN	18-MAY-25	One or more non-system datafiles are missing
---	------	------	-----------	--

RMAN> **list failure 2 detail;**

Database Role: PRIMARY

List of Database Failures

=====

Failure ID	Priority	Status	Time Detected	Summary
------------	----------	--------	---------------	---------

2	HIGH	OPEN	18-MAY-25	One or more non-system datafiles are missing
---	------	------	-----------	--

Impact: See impact for individual child failures

List of child failures for parent failure ID 2

Failure ID	Priority	Status	Time Detected	Summary
------------	----------	--------	---------------	---------

5745	HIGH	OPEN	18-MAY-25	Datafile 7: '/u01/app/oracle/oradata/ORCL/users01.dbf' is
------	------	------	-----------	---

missing

Impact: Some objects in tablespace USERS might be unavailable

-- 백업본 확인

RMAN> **list backup;**

-- 아카이브 로그 확인

RMAN> **list archivelog all;**

List of Archived Log Copies for database with db_unique_name ORCL

-- Recover

RMAN> **alter tablespace users offline immediate;**

Statement processed

RMAN> **restore tablespace users;**

Starting restore at 18-MAY-25

using channel ORA_DISK_1

channel ORA_DISK_1: starting datafile backup set restore

channel ORA_DISK_1: specifying datafile(s) to restore from backup set

channel ORA_DISK_1: restoring datafile 00007 to /u01/app/oracle/oradata/ORCL/users01.dbf

channel ORA_DISK_1: reading from backup piece
/home/oracle/backup/FRA/ORCL/backupset/2025_05_18/o1_mf_nnndf_TAG20250518T001132_n2k9s4d8_.bkp

channel ORA_DISK_1: piece
handle=/home/oracle/backup/FRA/ORCL/backupset/2025_05_18/o1_mf_nnndf_TAG20250518T001132_n2k9s4d8_.b
kp tag=TAG20250518T001132

channel ORA_DISK_1: restored backup piece 1

channel ORA_DISK_1: restore complete, elapsed time: 00:00:01

Finished restore at 18-MAY-25

RMAN> **recover tablespace users;**

Starting recover at 18-MAY-25

using channel ORA_DISK_1

starting media recovery

archived log for thread 1 with sequence 7 is already on disk as file /home/oracle/arch1/arch_1_7_1201218512.arc

archived log for thread 1 with sequence 8 is already on disk as file /home/oracle/arch1/arch_1_8_1201218512.arc

archived log for thread 1 with sequence 9 is already on disk as file /home/oracle/arch1/arch_1_9_1201218512.arc

archived log for thread 1 with sequence 10 is already on disk as file
/home/oracle/arch1/arch_1_10_1201218512.arc

archived log file name=/home/oracle/arch1/arch_1_7_1201218512.arc thread=1 sequence=7

archived log file name=/home/oracle/arch1/arch_1_8_1201218512.arc thread=1 sequence=8

media recovery complete, elapsed time: 00:00:00

Finished recover at 18-MAY-25

RMAN> alter tablespace users online;

Statement processed

RMAN> select count(*) from hr.loc1 ;

COUNT(*)

23

RMAN>

<<시나리오 3>> 일반 TS를 다른 위치에 복구

```
SYS@orcl> alter system switch logfile; <-- 여러번 수행
System altered.
```

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;
```

```
SYS@orcl> select sequence#, name, FIRST_CHANGE#, NEXT_CHANGE# from v$archived_log;
```

```
RMAN> report schema;
```

■ 복구 시나리오

```
SYS@orcl> create tablespace shtbs datafile '/home/oracle/oradata/shtbs01.dbf' size 10m;
Tablespace created. <-- Alert log 확인
```

```
SYS@orcl> create table hr.emp2 tablespace shtbs as select * from hr.employees;
Table created.
```

```
RMAN> report schema;
```

starting full resync of recovery catalog

full resync complete

Report of database schema for database with db_unique_name ORCL

List of Permanent Datafiles

=====

File	Size(MB)	Tablespace	RB segs	Datafile Name
1	910	SYSTEM	YES	/u01/app/oracle/oradata/ORCL/system01.dbf
3	600	SYSAUX	NO	/u01/app/oracle/oradata/ORCL/sysaux01.dbf
4	345	UNDOTBS1	YES	/u01/app/oracle/oradata/ORCL/undotbs01.dbf
5	10	SHTBS	NO	/home/oracle/oradata/shtbs01.dbf
7	5	USERS	NO	/u01/app/oracle/oradata/ORCL/users01.dbf

List of Temporary Files

=====

File	Size(MB)	Tablespace	Maxsize(MB)	Tempfile Name
1	32	TEMP	32767	/u01/app/oracle/oradata/ORCL/temp01.dbf

RMAN> **report need backup;**

RMAN retention policy will be applied to the command

RMAN retention policy is set to redundancy 1

Report of files with less than 1 redundant backups

File #bkps Name

```
-----
5      0      /home/oracle/oradata/shtbs01.dbf
```

RMAN> **backup tablespace shtbs;**

Starting backup at 18-MAY-25

using channel ORA_DISK_1

channel ORA_DISK_1: starting full datafile backup set

channel ORA_DISK_1: specifying datafile(s) in backup set

input datafile file number=00005 name=/home/oracle/oradata/shtbs01.dbf

channel ORA_DISK_1: starting piece 1 at 18-MAY-25

channel ORA_DISK_1: finished piece 1 at 18-MAY-25

piece

handle=/home/oracle/backup/FRA/ORCL/backupset/2025_05_18/o1_mf_nnndf_TAG20250518T005730_n2kdhbtpr.b

kp tag=TAG20250518T005730 comment=NONE

channel ORA_DISK_1: backup set complete, elapsed time: 00:00:01

Finished backup at 18-MAY-25

Starting Control File and SPFILE Autobackup at 18-MAY-25

piece handle=/home/oracle/backup/FRA/ORCL/autobackup/2025_05_18/o1_mf_s_1201395452_n2kdhdr7_.bkp

comment=NONE

Finished Control File and SPFILE Autobackup at 18-MAY-25

RMAN> **list backup;**

RMAN> **list backup of tablespace shtbs;** <---TS 단위 확인

List of Backup Sets

=====

BS Key	Type	LV	Size	Device	Type	Elapsed Time	Completion Time
--------	------	----	------	--------	------	--------------	-----------------

640	Full		1.09M	DISK		00:00:00	18-MAY-25
-----	------	--	-------	------	--	----------	-----------

BP Key: 643	Status: AVAILABLE	Compressed: NO	Tag: TAG20250518T005244
-------------	-------------------	----------------	-------------------------

Piece

Name:

/home/oracle/backup/FRA/ORCL/backupset/2025_05_18/o1_mf_nnndf_TAG20250518T005244_n2kd6dtx_.bkp

List of Datafiles in backup set 640

File	LV	Type	Ckp SCN	Ckp Time	Abs Fuz SCN	Sparse	Name
5		Full	3346207	18-MAY-25		NO	/home/oracle/oradata/shtbs01.dbf

BS Key	Type	LV	Size	Device	Type	Elapsed Time	Completion Time
710	Full		1.09M	DISK		00:00:00	18-MAY-25

BP Key: 713 Status: AVAILABLE Compressed: NO Tag: TAG20250518T005730

Piece

Name:

/home/oracle/backup/FRA/ORCL/backupset/2025_05_18/o1_mf_nnndf_TAG20250518T005730_n2kdhbtp_.bkp

List of Datafiles in backup set 710

File	LV	Type	Ckp SCN	Ckp Time	Abs Fuz SCN	Sparse	Name
5		Full	3346401	18-MAY-25		NO	/home/oracle/oradata/shtbs01.dbf

RMAN> list backup of datafile "/home/oracle/oradata/shtbs01.dbf"; <---DF 단위 확인

List of Backup Sets

=====

BS Key	Type	LV	Size	Device	Type	Elapsed Time	Completion Time
640	Full		1.09M	DISK		00:00:00	18-MAY-25

BP Key: 643 Status: AVAILABLE Compressed: NO Tag: TAG20250518T005244

Piece

Name:

/home/oracle/backup/FRA/ORCL/backupset/2025_05_18/o1_mf_nnndf_TAG20250518T005244_n2kd6dtx_.bkp

List of Datafiles in backup set 640

File	LV	Type	Ckp SCN	Ckp Time	Abs Fuz SCN	Sparse	Name
5		Full	3346207	18-MAY-25		NO	/home/oracle/oradata/shtbs01.dbf

BS Key	Type	LV	Size	Device	Type	Elapsed Time	Completion Time
710	Full		1.09M	DISK		00:00:00	18-MAY-25

BP Key: 713 Status: AVAILABLE Compressed: NO Tag: TAG20250518T005730

Piece

Name:

/home/oracle/backup/FRA/ORCL/backupset/2025_05_18/o1_mf_nnndf_TAG20250518T005730_n2kdhbtp_.bkp

List of Datafiles in backup set 710

File	LV	Type	Ckp SCN	Ckp Time	Abs Fuz SCN	Sparse	Name
5		Full	3346401	18-MAY-25		NO	/home/oracle/oradata/shtbs01.dbf

- 장애 발생

```
RMAN> alter system switch logfile;      <---- 여러번 수행
Statement processed
```

```
SYS@orcl> ! rm /home/oracle/oradata/shtbs01.dbf
```

```
SYS@orcl> !ls /home/oracle/oradata/shtbs01.dbf
```

ls: cannot access /home/oracle/oradata/shtbs01.dbf: No such file or directory

약간이 시간이 필요.

```
SYS@orcl> alter system checkpoint;
alter system checkpoint
*
```

ERROR at line 1:

ORA-03135: connection lost contact

Process ID: 24972

Session ID: 79 Serial number: 9475

-- Alert log 확인

2025-05-18T01:06:12.067261+09:00

Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl_mz00_25716.trc:

ORA-01110: data file 5: '/home/oracle/oradata/shtbs01.dbf'

ORA-01565: error in identifying file '/home/oracle/oradata/shtbs01.dbf'

ORA-27037: unable to obtain file status

Linux-x86_64 Error: 2: No such file or directory

Additional information: 7

```
SYS@orcl> conn /as sysdba
```

Connected to an idle instance.

```
SYS@orcl> startup
```

ORACLE instance started.

Total System Global Area 1140849904 bytes

Fixed Size 8895728 bytes

Variable Size 822083584 bytes

Database Buffers 301989888 bytes

Redo Buffers 7880704 bytes

Database mounted.

ORA-01157: cannot identify/lock data file 5 - see DBWR trace file

ORA-01110: data file 5: '/home/oracle/oradata/shtbs01.dbf'

```
SYS@orcl> select status from v$instance;
```

```
STATUS
```

```
-----
```

```
MOUNTED
```

```
- rman 재 접속
```

```
[oracle@oel7v9r1 admin]$ rman target "'sys/oracle@t1 as sysdba'" catalog ruser/ruser@c1
```

```
RMAN> list failure;
```

```
Database Role: PRIMARY
```

```
List of Database Failures
```

```
=====
```

```
Failure ID Priority Status Time Detected Summary
```

```
-----
```

2	HIGH	OPEN	18-MAY-25	One or more non-system datafiles are missing
5762	HIGH	OPEN	18-MAY-25	Tablespace 4: 'USERS' is offline

```
RMAN> list failure 5762 detail;
```

```
Database Role: PRIMARY
```

```
List of Database Failures
```

```
=====
```

```
Failure ID Priority Status Time Detected Summary
```

```
-----
```

5762	HIGH	OPEN	18-MAY-25	Tablespace 4: 'USERS' is offline
------	------	------	-----------	----------------------------------

Impact: Some objects in tablespace USERS might be unavailable

```
RMAN> list failure 2 detail;
```

```
Database Role: PRIMARY
```

```
List of Database Failures
```

```
=====
```

```
Failure ID Priority Status Time Detected Summary
```

```
-----
```

2	HIGH	OPEN	18-MAY-25	One or more non-system datafiles are missing
---	------	------	-----------	--

Impact: See impact for individual child failures

```
List of child failures for parent failure ID 2
```

```
Failure ID Priority Status Time Detected Summary
```

```
-----
```

5885	HIGH	OPEN	18-MAY-25	Datafile 5: '/home/oracle/oradata/shtbs01.dbf' is missing
------	------	------	-----------	---

Impact: Some objects in tablespace SHTBS might be unavailable

-- Recover

RMAN> **restore datafile 5;**

Starting restore at 18-MAY-25

using channel ORA_DISK_1

channel ORA_DISK_1: starting datafile backup set restore

channel ORA_DISK_1: specifying datafile(s) to restore from backup set

channel ORA_DISK_1: restoring datafile 00005 to /home/oracle/oradata/shtbs01.dbf

channel ORA_DISK_1: reading from backup piece
/home/oracle/backup/FRA/ORCL/backupset/2025_05_18/o1_mf_nnndf_TAG20250518T005730_n2kdhbtb_bkp

channel ORA_DISK_1: piece
handle=/home/oracle/backup/FRA/ORCL/backupset/2025_05_18/o1_mf_nnndf_TAG20250518T005730_n2kdhbtb_bkp tag=TAG20250518T005730

channel ORA_DISK_1: restored backup piece 1

channel ORA_DISK_1: restore complete, elapsed time: 00:00:01

Finished restore at 18-MAY-25

RMAN> **recover datafile 5;**

Starting recover at 18-MAY-25

using channel ORA_DISK_1

starting media recovery

media recovery complete, elapsed time: 00:00:00

Finished recover at 18-MAY-25

RMAN> **alter database datafile 5 online;**

Statement processed

RMAN> **report schema;**

Report of database schema for database with db_unique_name ORCL

List of Permanent Datafiles

=====

File	Size(MB)	Tablespace	RB segs	Datafile Name
------	----------	------------	---------	---------------

1	910	SYSTEM	YES	/u01/app/oracle/oradata/ORCL/system01.dbf
---	-----	--------	-----	---

3	600	SYSAUX	NO	/u01/app/oracle/oradata/ORCL/sysaux01.dbf
4	345	UNDOTBS1	YES	/u01/app/oracle/oradata/ORCL/undotbs01.dbf
5	10	SHTBS	NO	/home/oracle/oradata/shtbs01.dbf
7	5	USERS	NO	/u01/app/oracle/oradata/ORCL/users01.dbf

List of Temporary Files

=====

File	Size(MB)	Tablespace	Maxsize(MB)	Tempfile Name
1	32	TEMP	32767	/u01/app/oracle/oradata/ORCL/temp01.dbf

- 장애 발생(한번 더)

```
SYS@orcl> update hr.emp2 set salary=5000;
107 rows updated.
```

```
SYS@orcl> insert into hr.emp2 select * from hr.emp2;
107 rows created.
```

```
SYS@orcl> commit;
```

```
RMAN> backup tablespace shtbs;
```

Starting backup at 18-MAY-25

using channel ORA_DISK_1

channel ORA_DISK_1: starting full datafile backup set

channel ORA_DISK_1: specifying datafile(s) in backup set

input datafile file number=00005 name=/home/oracle/oradata/shtbs01.dbf

channel ORA_DISK_1: starting piece 1 at 18-MAY-25

channel ORA_DISK_1: finished piece 1 at 18-MAY-25

piece

handle=/home/oracle/backup/FRA/ORCL/backupset/2025_05_18/o1_mf_nnndf_TAG20250518T013231_n2kgjzld_.bk

p tag=TAG20250518T013231 comment=NONE

channel ORA_DISK_1: backup set complete, elapsed time: 00:00:01

Finished backup at 18-MAY-25

Starting Control File and SPFILE Autobackup at 18-MAY-25

piece handle=/home/oracle/backup/FRA/ORCL/autobackup/2025_05_18/o1_mf_s_1201397553_n2kgk1f6_.bkp

comment=NONE

Finished Control File and SPFILE Autobackup at 18-MAY-25


```
SYS@orcl> ! rm /home/oracle/oradata/shtbs01.dbf
```

```
SYS@orcl> !ls /home/oracle/oradata/shtbs01.dbf
```

```
ls: cannot access /home/oracle/oradata/shtbs01.dbf: No such file or directory
```

```
-Alert log
```

```
ORA-01110: data file 5: '/home/oracle/oradata/shtbs01.dbf'
```

```
ORA-01565: error in identifying file '/home/oracle/oradata/shtbs01.dbf'
```

```
ORA-27037: unable to obtain file status
```

```
Linux-x86_64 Error: 2: No such file or directory
```

```
SYS@orcl> conn /as sysdba
```

```
Connected.
```

```
SYS@orcl> shut abort
```

```
ORACLE instance shut down.
```

```
SYS@orcl> startup
```

```
ORACLE instance started.
```

```
Total System Global Area 1140849904 bytes
```

```
Fixed Size 8895728 bytes
```

```
Variable Size 822083584 bytes
```

```
Database Buffers 301989888 bytes
```

```
Redo Buffers 7880704 bytes
```

```
Database mounted.
```

```
ORA-01157: cannot identify/lock data file 5 - see DBWR trace file
```

```
ORA-01110: data file 5: '/home/oracle/oradata/shtbs01.dbf'
```

```
SYS@orcl> select status from v$instance;
```

```
STATUS
```

```
-----
```

```
MOUNTED
```

```
- rman 재 접속
```

```
[oracle@oel7v9r1 admin]$ rman target "sys/oracle@t1 as sysdba" catalog ruser/ruser@c1
```

```
RMAN> select * from v$recover_file;
```

FILE#	ONLINE	ONLINE_ ERROR	CHANGE#	TIME	CON_ID
5	ONLINE	ONLINE	FILE NOT FOUND	0	0

RMAN> **list failure;**

Database Role: PRIMARY

List of Database Failures

=====

Failure ID	Priority	Status	Time Detected	Summary
5942	HIGH	OPEN	18-MAY-25	One or more non-system datafiles are corrupt
2	HIGH	OPEN	18-MAY-25	One or more non-system datafiles are missing
5762	HIGH	OPEN	18-MAY-25	Tablespace 4: 'USERS' is offline

RMAN> **list failure 5942 detail;**

Database Role: PRIMARY

List of Database Failures

=====

Failure ID	Priority	Status	Time Detected	Summary
5942	HIGH	OPEN	18-MAY-25	One or more non-system datafiles are corrupt

Impact: See impact for individual child failures

List of child failures for parent failure ID 5942

Failure ID	Priority	Status	Time Detected	Summary
5945	HIGH	OPEN	18-MAY-25	Datafile 5: '/home/oracle/oradata/shtbs01.dbf' is corrupt

Impact: Some objects in tablespace SHTBS might be unavailable

RMAN> **alter database datafile 5 offline;**

Statement processed

RMAN> **alter database open;**

Statement processed

RMAN> **select status from v\$instance;**

STATUS

OPEN

RMAN> **select * from hr.employees;**

COUNT(*)

107

```

RMAN> select count(*) from hr.emp2;
RMAN-00571: =====
RMAN-00569: ===== ERROR MESSAGE STACK FOLLOWS =====
RMAN-00571: =====
RMAN-03002: failure of sql statement command at 05/18/2025 01:52:57
ORA-00376: file 5 cannot be read at this time
ORA-01110: data file 5: '/home/oracle/oradata/shtbs01.dbf'

```

- 새로운 위치로 이동 (RMAN 사용시)

```

1. set newname for datafile 파일번호 '새로운위치'; ( run 모드로 수행)
set newname for datafile 5 '/u01/app/oracle/oradata/ORCL/shtbs01.dbf';
2. restore datafile 파일번호;
restore datafile 5;
3. switch datafile 파일번호;
-- switch는 rename과 같은 작업
-- 해당 명령어는 run 모드로 수행
switch datafile 4;
4. recover datafile 파일번호;
recover datafile 4;
5. alter database datafile 파일번호 4 online;
alter database datafile 4 online;

```

- RUN 모드 수행

```

RMAN> run{
2> set newname for datafile 5 to '/u01/app/oracle/oradata/ORCL/shtbs01.dbf';
3> restore datafile 5;
4> switch datafile 5;
5> recover datafile 5;
6> alter database datafile 5 online;
7> }

```

executing command: SET NEWNAME

Starting restore at 18-MAY-25

allocated channel: ORA_DISK_1

channel ORA_DISK_1: SID=73 device type=DISK

channel ORA_DISK_1: starting datafile backup set restore

channel ORA_DISK_1: specifying datafile(s) to restore from backup set

channel ORA_DISK_1: restoring datafile 00005 to /u01/app/oracle/oradata/ORCL/shtbs01.dbf

channel	ORA_DISK_1:	reading	from	backup	piece
---------	-------------	---------	------	--------	-------

```
/home/oracle/backup/FRA/ORCL/backupset/2025_05_18/o1_mf_nnndf_TAG20250518T013231_n2kgjzld_.bkp
channel ORA_DISK_1: piece
handle=/home/oracle/backup/FRA/ORCL/backupset/2025_05_18/o1_mf_nnndf_TAG20250518T013231_n2kgjzld_.bk
p tag=TAG20250518T013231
channel ORA_DISK_1: restored backup piece 1
channel ORA_DISK_1: restore complete, elapsed time: 00:00:01
Finished restore at 18-MAY-25
```

```
datafile 5 switched to datafile copy
input datafile copy RECID=2 STAMP=1201399201 file name=/u01/app/oracle/oradata/ORCL/shtbs01.dbf
starting full resync of recovery catalog
full resync complete
```

```
Starting recover at 18-MAY-25
using channel ORA_DISK_1
```

```
starting media recovery
```

```
archived log for thread 1 with sequence 20 is already on disk as file
/home/oracle/arch1/arch_1_20_1201218512.arc
archived log for thread 1 with sequence 21 is already on disk as file
/home/oracle/arch1/arch_1_21_1201218512.arc
archived log for thread 1 with sequence 22 is already on disk as file
/home/oracle/arch1/arch_1_22_1201218512.arc
archived log for thread 1 with sequence 23 is already on disk as file
/home/oracle/arch1/arch_1_23_1201218512.arc
archived log for thread 1 with sequence 24 is already on disk as file
/home/oracle/arch1/arch_1_24_1201218512.arc
archived log file name=/home/oracle/arch1/arch_1_20_1201218512.arc thread=1 sequence=20
archived log file name=/home/oracle/arch1/arch_1_21_1201218512.arc thread=1 sequence=21
archived log file name=/home/oracle/arch1/arch_1_22_1201218512.arc thread=1 sequence=22
media recovery complete, elapsed time: 00:00:00
Finished recover at 18-MAY-25
```

```
Statement processed
```

```
RMAN> select file#, name from v$datafile;
```

FILE#	NAME
1	/u01/app/oracle/oradata/ORCL/system01.dbf

```

3 /u01/app/oracle/oradata/ORCL/sysaux01.dbf
4 /u01/app/oracle/oradata/ORCL/undotbs01.dbf
5 /u01/app/oracle/oradata/ORCL/shtbs01.dbf
7 /u01/app/oracle/oradata/ORCL/users01.dbf

```

```

RMAN> select count(*) from hr.emp2;
COUNT(*)

```

```

-----
1712

```

```

RMAN> report obsolete;

```

RMAN retention policy will be applied to the command

RMAN retention policy is set to redundancy 1

Report of obsolete backups and copies

Type	Key	Completion Time	Filename/Handle
Archive Log	194	15-MAY-25	/home/oracle/arch1/arch_1_13_1201040433.arc
Archive Log	195	15-MAY-25	/home/oracle/arch2/arch_1_13_1201040433.arc
Archive Log	196	15-MAY-25	/home/oracle/arch1/arch_1_14_1201040433.arc
Archive Log	197	15-MAY-25	/home/oracle/arch2/arch_1_14_1201040433.arc
Archive Log	198	15-MAY-25	/home/oracle/arch1/arch_1_12_1201040433.arc
Archive Log	199	15-MAY-25	/home/oracle/arch2/arch_1_12_1201040433.arc
Archive Log	200	17-MAY-25	/home/oracle/arch1/arch_1_1_1201218512.arc
Archive Log	201	17-MAY-25	/home/oracle/arch2/arch_1_1_1201218512.arc
Archive Log	202	17-MAY-25	/home/oracle/arch1/arch_1_2_1201218512.arc
Archive Log	203	17-MAY-25	/home/oracle/arch2/arch_1_2_1201218512.arc
Archive Log	204	17-MAY-25	/home/oracle/arch1/arch_1_3_1201218512.arc
Archive Log	205	17-MAY-25	/home/oracle/arch2/arch_1_3_1201218512.arc
Archive Log	206	17-MAY-25	/home/oracle/arch1/arch_1_4_1201218512.arc
Archive Log	207	17-MAY-25	/home/oracle/arch2/arch_1_4_1201218512.arc
Archive Log	308	17-MAY-25	/home/oracle/arch1/arch_1_5_1201218512.arc
Archive Log	309	17-MAY-25	/home/oracle/arch2/arch_1_5_1201218512.arc
Archive Log	413	17-MAY-25	/home/oracle/arch1/arch_1_6_1201218512.arc
Archive Log	414	17-MAY-25	/home/oracle/arch2/arch_1_6_1201218512.arc
Backup Set	532	18-MAY-25	
Backup	Piece		537
/home/oracle/backup/FRA/ORCL/autobackup/2025_05_18/o1_mf_s_1201392700_n2k9sdbw_.bkp			
Backup Set	606	18-MAY-25	
Backup	Piece		609
/home/oracle/backup/FRA/ORCL/autobackup/2025_05_18/o1_mf_s_1201394763_n2kcsvw8_.bkp			

```

Backup Set          640    18-MAY-25
  Backup      Piece                      643                      18-MAY-25
/home/oracle/backup/FRA/ORCL/backupset/2025_05_18/o1_mf_nnndf_TAG20250518T005244_n2kd6dtx_.bkp
Backup Set          673    18-MAY-25
  Backup      Piece                      675                      18-MAY-25
/home/oracle/backup/FRA/ORCL/autobackup/2025_05_18/o1_mf_s_1201395166_n2kd6gwq_.bkp
Backup Set          710    18-MAY-25
  Backup      Piece                      713                      18-MAY-25
/home/oracle/backup/FRA/ORCL/backupset/2025_05_18/o1_mf_nnndf_TAG20250518T005730_n2kdhbtp_.bkp
Backup Set          720    18-MAY-25
  Backup      Piece                      722                      18-MAY-25
/home/oracle/backup/FRA/ORCL/autobackup/2025_05_18/o1_mf_s_1201395452_n2kdhdr7_.bkp
Backup Set          759    18-MAY-25
  Backup      Piece                      762                      18-MAY-25
/home/oracle/backup/FRA/ORCL/autobackup/2025_05_18/o1_mf_s_1201397007_n2kfzzc4_.bkp

```

=====

- 정책상 필요없는 백업정보, 아카이브 정보 삭제

delete obsolete;

- 물리적인 아카이브 파일과 컨트롤 파일이 가지고 있는 아카이브 파일을 체크

crosscheck archivelog all;

- 크로스 체크한 후 문제되는 정보를 expired 된 아카이브 파일 확인

list expired archivelog all;

- expired 된 아카이브 파일 삭제

delete expired archivelog all;

=====

RMAN> crosscheck archivelog all;

released channel: ORA_DISK_1

allocated channel: ORA_DISK_1

channel ORA_DISK_1: SID=73 device type=DISK

validation succeeded for archived log

archived log file name=/home/oracle/arch2/arch_1_12_1201040433.arc RECID=92 STAMP=1201218512

validation succeeded for archived log

RMAN> list expired archivelog all;

RMAN> delete obsolete;

RMAN retention policy will be applied to the command

RMAN retention policy is set to redundancy 1

using channel ORA_DISK_1

Deleting the following obsolete backups and copies:

Type	Key	Completion Time	Filename/Handle
Archive Log	308	17-MAY-25	/home/oracle/arch1/arch_1_5_1201218512.arc
Archive Log	309	17-MAY-25	/home/oracle/arch2/arch_1_5_1201218512.arc
Archive Log	413	17-MAY-25	/home/oracle/arch1/arch_1_6_1201218512.arc
Archive Log	414	17-MAY-25	/home/oracle/arch2/arch_1_6_1201218512.arc
Backup Set	532	18-MAY-25	
Backup Piece			537 18-MAY-25
			/home/oracle/backup/FRA/ORCL/autobackup/2025_05_18/o1_mf_s_1201392700_n2k9sdbw_.bkp
Backup Set	606	18-MAY-25	
Backup Piece			609 18-MAY-25
			/home/oracle/backup/FRA/ORCL/autobackup/2025_05_18/o1_mf_s_1201394763_n2kcsvw8_.bkp
Backup Set	640	18-MAY-25	
Backup Piece			643 18-MAY-25
			/home/oracle/backup/FRA/ORCL/backupset/2025_05_18/o1_mf_nnndf_TAG20250518T005244_n2kd6dtx_.bkp
Backup Set	673	18-MAY-25	
Backup Piece			675 18-MAY-25
			/home/oracle/backup/FRA/ORCL/autobackup/2025_05_18/o1_mf_s_1201395166_n2kd6gwq_.bkp
Backup Set	710	18-MAY-25	
Backup Piece			713 18-MAY-25
			/home/oracle/backup/FRA/ORCL/backupset/2025_05_18/o1_mf_nnndf_TAG20250518T005730_n2kdhbtp_.bkp
Backup Set	720	18-MAY-25	
Backup Piece			722 18-MAY-25
			/home/oracle/backup/FRA/ORCL/autobackup/2025_05_18/o1_mf_s_1201395452_n2kdhdr7_.bkp
Backup Set	759	18-MAY-25	
Backup Piece			762 18-MAY-25
			/home/oracle/backup/FRA/ORCL/autobackup/2025_05_18/o1_mf_s_1201397007_n2kfzzc4_.bkp
Backup Set	823	18-MAY-25	
Backup Piece			825 18-MAY-25
			/home/oracle/backup/FRA/ORCL/autobackup/2025_05_18/o1_mf_s_1201397553_n2kgk1f6_.bkp

Do you really want to delete the above objects (enter YES or NO)? YES

RMAN> list backup;

List of Backup Sets

=====

BS Key	Type	LV Size	Device Type	Elapsed Time	Completion Time
--------	------	---------	-------------	--------------	-----------------

516	Full	1.21G	DISK	00:00:03	18-MAY-25
-----	------	-------	------	----------	-----------

BP Key: 517 Status: AVAILABLE Compressed: NO Tag: TAG20250518T001132

Piece

Name:

/home/oracle/backup/FRA/ORCL/backupset/2025_05_18/o1_mf_nnndf_TAG20250518T001132_n2k9s4d8_.bkp

List of Datafiles in backup set 516

File	LV Type	Ckp SCN	Ckp Time	Abs Fuz SCN	Sparse	Name
------	---------	---------	----------	-------------	--------	------

1	Full	3342795	18-MAY-25		NO	/u01/app/oracle/oradata/ORCL/system01.dbf
---	------	---------	-----------	--	----	---

3	Full	3342795	18-MAY-25		NO	/u01/app/oracle/oradata/ORCL/sysaux01.dbf
---	------	---------	-----------	--	----	---

4	Full	3342795	18-MAY-25		NO	/u01/app/oracle/oradata/ORCL/undotbs01.dbf
---	------	---------	-----------	--	----	--

7	Full	3342795	18-MAY-25		NO	/u01/app/oracle/oradata/ORCL/users01.dbf
---	------	---------	-----------	--	----	--

BS Key	Type	LV Size	Device Type	Elapsed Time	Completion Time
--------	------	---------	-------------	--------------	-----------------

802	Full	1.09M	DISK	00:00:00	18-MAY-25
-----	------	-------	------	----------	-----------

BP Key: 805 Status: AVAILABLE Compressed: NO Tag: TAG20250518T013231

Piece

Name:

/home/oracle/backup/FRA/ORCL/backupset/2025_05_18/o1_mf_nnndf_TAG20250518T013231_n2kgjzld_.bkp

List of Datafiles in backup set 802

File	LV Type	Ckp SCN	Ckp Time	Abs Fuz SCN	Sparse	Name
------	---------	---------	----------	-------------	--------	------

5	Full	3449596	18-MAY-25		NO	/u01/app/oracle/oradata/ORCL/shtbs01.dbf
---	------	---------	-----------	--	----	--

BS Key	Type	LV Size	Device Type	Elapsed Time	Completion Time
--------	------	---------	-------------	--------------	-----------------

919	Full	10.30M	DISK	00:00:00	18-MAY-25
-----	------	--------	------	----------	-----------

BP Key: 922 Status: AVAILABLE Compressed: NO Tag: TAG20250518T020922

Piece

Name:

/home/oracle/backup/FRA/ORCL/autobackup/2025_05_18/o1_mf_s_1201399762_n2kjp2wj_.bkp

SPFILE Included: Modification time: 18-MAY-25

SPFILE db_unique_name: ORCL

Control File Included: Ckp SCN: 3553770 Ckp time: 18-MAY-25

<<시나리오 4>> System DF 손상되었을 경우 복구

```
RMAN> backup database;
```

```
RMAN> list backup;
```

```

RMAN> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;

```

FILE#	TBS_NAME	FILE_NAME	CHECKPOINT_CHANGE#	STATUS
3	SYSAUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	3555016	ONLINE
1	SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf	3555016	SYSTEM
4	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	3555016	ONLINE
7	USERS	/u01/app/oracle/oradata/ORCL/users01.dbf	3555016	ONLINE
5	SHTBS	/u01/app/oracle/oradata/ORCL/shtbs01.dbf	3555016	ONLINE

```
RMAN> report schema;
```

```

RMAN> select sequence#, name, FIRST_CHANGE#, NEXT_CHANGE# from v$archived_log;

```

```

RMAN> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;

```

GROUP#	SEQUENCE#	MEMBER	

MB STATUS	ARC	FIRST_CHANGE#	NEXT_CHANGE#

1	25	/u01/app/oracle/oradata/ORCL/redo01.log	
10 CURRENT	NO	3550229	184467440737
2	23	/u01/app/oracle/oradata/ORCL/redo02.log	
10 INACTIVE	YES	3449760	3449763
3	24	/u01/app/oracle/oradata/ORCL/redo03.log	
10 INACTIVE	YES	3449763	3550229

■ 복구 시나리오

- 장애 발생

```
SYS@orcl> select status from v$instance;
```

STATUS

OPEN

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/system01.dbf
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/system01.dbf
```

```
ls: cannot access /u01/app/oracle/oradata/ORCL/system01.dbf: No such file or directory
```

```
RMAN> alter system checkpoint;
```

```
RMAN-00571: =====
```

```
RMAN-00569: ===== ERROR MESSAGE STACK FOLLOWS =====
```

```
RMAN-00571: =====
```

```
RMAN-00601: fatal error in recovery manager
```

```
RMAN-03004: fatal error during execution of command
```

```
ORA-01092: ORACLE instance terminated. Disconnection forced
```

```
RMAN-03002: failure of sql statement command at 05/18/2025 02:27:38
```

```
ORA-03113: end-of-file on communication channel
```

```
Process ID: 28539
```

```
Session ID: 44 Serial number: 51650
```

- Alert log file

```
2025-05-18T02:27:37.174332+09:00
```

```
Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl_mz00_31341.trc:
```

```
ORA-01110: data file 1: '/u01/app/oracle/oradata/ORCL/system01.dbf'
```

```
ORA-01565: error in identifying file '/u01/app/oracle/oradata/ORCL/system01.dbf'
```

```
ORA-27037: unable to obtain file status
```

```
Linux-x86_64 Error: 2: No such file or directory
```

```
----
```

```
2025-05-18T02:27:39.253174+09:00
```

```
Dumping diagnostic data in directory=[cdmp_20250518022738], requested by (instance=1, osid=28433 (CKPT)),  
summary=[abnormal instance termination].
```

```
2025-05-18T02:27:40.421774+09:00
```

```
Instance terminated by USER, pid = 28433
```

```
[oracle@oel7v9r1 admin]$ rman target "'sys/oracle@t1 as sysdba'" catalog ruser/ruser@c1
```

- Recovery

```
RMAN> startup
```

```
Oracle instance started
```

```
database mounted
```

```
RMAN-00571: =====
```

```

RMAN-00569: ===== ERROR MESSAGE STACK FOLLOWS =====
RMAN-00571: =====
RMAN-03002: failure of startup command at 05/18/2025 02:30:46
ORA-01157: cannot identify/lock data file 1 - see DBWR trace file
ORA-01110: data file 1: '/u01/app/oracle/oradata/ORCL/system01.dbf'

```

RMAN> **list failure;**

Database Role: PRIMARY

List of Database Failures

=====

Failure ID	Priority	Status	Time Detected	Summary
6162	CRITICAL	OPEN	18-MAY-25	System datafile 1: '/u01/app/oracle/oradata/ORCL/system01.dbf' is missing
5762	HIGH	OPEN	18-MAY-25	Tablespace 4: 'USERS' is offline

RMAN> list failure 6162 detail;

Database Role: PRIMARY

List of Database Failures

=====

Failure ID	Priority	Status	Time Detected	Summary
6162	CRITICAL	OPEN	18-MAY-25	System datafile 1: '/u01/app/oracle/oradata/ORCL/system01.dbf' is missing

Impact: Database cannot be opened

RMAN> **select status from v\$instance;**

STATUS

MOUNTED

RMAN> **restore datafile 1;**

Starting restore at 18-MAY-25

allocated channel: ORA_DISK_1

channel ORA_DISK_1: SID=46 device type=DISK

channel ORA_DISK_1: starting datafile backup set restore

channel ORA_DISK_1: specifying datafile(s) to restore from backup set

channel ORA_DISK_1: restoring datafile 00001 to /u01/app/oracle/oradata/ORCL/system01.dbf

channel ORA_DISK_1: reading from backup piece
/home/oracle/backup/FRA/ORCL/backupset/2025_05_18/o1_mf_nnndf_TAG20250518T021707_n2kk4mp1_.bkp

channel ORA_DISK_1: piece
handle=/home/oracle/backup/FRA/ORCL/backupset/2025_05_18/o1_mf_nnndf_TAG20250518T021707_n2kk4mp1_.
bkp tag=TAG20250518T021707

channel ORA_DISK_1: restored backup piece 1

channel ORA_DISK_1: restore complete, elapsed time: 00:00:03

Finished restore at 18-MAY-25

RMAN> recover datafile 1;

Starting recover at 18-MAY-25

using channel ORA_DISK_1

starting media recovery

media recovery complete, elapsed time: 00:00:00

Finished recover at 18-MAY-25

RMAN> alter database open;

Statement processed

RMAN> list failure;

Database Role: PRIMARY

no failures found that match specification

<<시나리오 5>> 모든 DFs 손상되었을 경우 복구

```
SYS@orcl> select status from v$instance;
```

```
STATUS
```

```
-----
```

```
OPEN
```

```
SYS@orcl> alter system switch logfile; <--- 여러번 수행
```

```
System altered.
```

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,  
v$tablespace b where a.ts# = b.ts#;
```

■ 복구 시나리오

```
SYS@orcl> select sequence#, name, FIRST_CHANGE#, NEXT_CHANGE# from v$archived_log;
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/*.dbf
```

```
/u01/app/oracle/oradata/ORCL/shtbs01.dbf
```

```
/u01/app/oracle/oradata/ORCL/sysaux01.dbf
```

```
/u01/app/oracle/oradata/ORCL/system01.dbf
```

```
/u01/app/oracle/oradata/ORCL/temp01.dbf
```

```
/u01/app/oracle/oradata/ORCL/undotbs01.dbf
```

```
/u01/app/oracle/oradata/ORCL/users01.dbf
```

- 장애 발생

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/*.dbf
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/*.dbf
```

```
ls: cannot access /u01/app/oracle/oradata/ORCL/*.dbf: No such file or directory
```

```
SYS@orcl> alter system checkpoint;
```

- Alert log file

```
2025-05-18T02:42:48.544747+09:00
```

```
Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl_mz00_32559.trc:
```

```
ORA-01110: data file 1: '/u01/app/oracle/oradata/ORCL/system01.dbf'
```

```
ORA-01565: error in identifying file '/u01/app/oracle/oradata/ORCL/system01.dbf'
```

```
ORA-27037: unable to obtain file status
```

```
Linux-x86_64 Error: 2: No such file or directory
```

```
Additional information: 7
```

```
2025-05-18T02:42:48.666403+09:00
```

```
Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl_mz00_32559.trc:
```

ORA-01110: data file 3: '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'
ORA-01565: error in identifying file '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'
ORA-27037: unable to obtain file status
Linux-x86_64 Error: 2: No such file or directory
Additional information: 7
2025-05-18T02:42:48.712656+09:00
Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl_smon_31588.trc:
ORA-00604: error occurred at recursive SQL level 1
ORA-01116: error in opening database file 3
ORA-01110: data file 3: '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'
ORA-27041: unable to open file
Linux-x86_64 Error: 2: No such file or directory
Additional information: 3
2025-05-18T02:42:48.803888+09:00
Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl_mz00_32559.trc:
ORA-01110: data file 4: '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'
ORA-01565: error in identifying file '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'
ORA-27037: unable to obtain file status
Linux-x86_64 Error: 2: No such file or directory
Additional information: 7
2025-05-18T02:42:48.912174+09:00
Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl_mz00_32559.trc:
ORA-01110: data file 5: '/u01/app/oracle/oradata/ORCL/shtbs01.dbf'
ORA-01565: error in identifying file '/u01/app/oracle/oradata/ORCL/shtbs01.dbf'
ORA-27037: unable to obtain file status
Linux-x86_64 Error: 2: No such file or directory
Additional information: 7
2025-05-18T02:42:49.018954+09:00
Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl_mz00_32559.trc:
ORA-01110: data file 7: '/u01/app/oracle/oradata/ORCL/users01.dbf'
ORA-01565: error in identifying file '/u01/app/oracle/oradata/ORCL/users01.dbf'
ORA-27037: unable to obtain file status
Linux-x86_64 Error: 2: No such file or directory
Additional information: 7
Checker run found 5 new persistent data failures

SYS@orcl> shut abort
ORACLE instance shut down.
SYS@orcl>

```
[oracle@oel7v9r1 admin]$ rman target "'sys/oracle@t1 as sysdba'" catalog ruser/ruser@c1
```

```
RMAN> startup mount
```

```
RMAN> list failure;
```

Database Role: PRIMARY

List of Database Failures

=====

Failure ID	Priority	Status	Time Detected	Summary
------------	----------	--------	---------------	---------

6242	CRITICAL	OPEN	18-MAY-25	System datafile 1: '/u01/app/oracle/oradata/ORCL/system01.dbf' is missing
------	----------	------	-----------	---

2	HIGH	OPEN	18-MAY-25	One or more non-system datafiles are missing
---	------	------	-----------	--

```
RMAN> list failure 2 detail;
```

Database Role: PRIMARY

List of Database Failures

=====

Failure ID	Priority	Status	Time Detected	Summary
------------	----------	--------	---------------	---------

2	HIGH	OPEN	18-MAY-25	One or more non-system datafiles are missing
---	------	------	-----------	--

Impact: See impact for individual child failures

List of child failures for parent failure ID 2

Failure ID	Priority	Status	Time Detected	Summary
------------	----------	--------	---------------	---------

6266	HIGH	OPEN	18-MAY-25	Datafile 7: '/u01/app/oracle/oradata/ORCL/users01.dbf' is missing
------	------	------	-----------	---

Impact: Some objects in tablespace USERS might be unavailable

6260	HIGH	OPEN	18-MAY-25	Datafile 5: '/u01/app/oracle/oradata/ORCL/shtbs01.dbf' is missing
------	------	------	-----------	---

Impact: Some objects in tablespace SHTBS might be unavailable

6254	HIGH	OPEN	18-MAY-25	Datafile 4: '/u01/app/oracle/oradata/ORCL/undotbs01.dbf' is missing
------	------	------	-----------	---

Impact: Some objects in tablespace UNDOTBS1 might be unavailable

6248	HIGH	OPEN	18-MAY-25	Datafile 3: '/u01/app/oracle/oradata/ORCL/sysaux01.dbf' is missing
------	------	------	-----------	--

Impact: Some objects in tablespace SYSAUX might be unavailable

RMAN> **select * from v\$recover_file;**

FILE#	ONLINE	ONLINE_ ERROR	CHANGE#	TIME	CON_ID
1	ONLINE	ONLINE	FILE NOT FOUND	0	0
3	ONLINE	ONLINE	FILE NOT FOUND	0	0
4	ONLINE	ONLINE	FILE NOT FOUND	0	0
5	ONLINE	ONLINE	FILE NOT FOUND	0	0
7	ONLINE	ONLINE	FILE NOT FOUND	0	0

- Recovery

RMAN> **restore database;**

Starting restore at 18-MAY-25

allocated channel: ORA_DISK_1

channel ORA_DISK_1: SID=45 device type=DISK

channel ORA_DISK_1: starting datafile backup set restore

channel ORA_DISK_1: specifying datafile(s) to restore from backup set

channel ORA_DISK_1: restoring datafile 00001 to /u01/app/oracle/oradata/ORCL/system01.dbf

channel ORA_DISK_1: restoring datafile 00003 to /u01/app/oracle/oradata/ORCL/sysaux01.dbf

channel ORA_DISK_1: restoring datafile 00004 to /u01/app/oracle/oradata/ORCL/undotbs01.dbf

channel ORA_DISK_1: restoring datafile 00005 to /u01/app/oracle/oradata/ORCL/shtbs01.dbf

channel ORA_DISK_1: restoring datafile 00007 to /u01/app/oracle/oradata/ORCL/users01.dbf

channel ORA_DISK_1: reading from backup piece
/home/oracle/backup/FRA/ORCL/backupset/2025_05_18/o1_mf_nnndf_TAG20250518T021707_n2kk4mp1_.bkp

channel ORA_DISK_1: piece
handle=/home/oracle/backup/FRA/ORCL/backupset/2025_05_18/o1_mf_nnndf_TAG20250518T021707_n2kk4mp1_.
bkp tag=TAG20250518T021707

channel ORA_DISK_1: restored backup piece 1

channel ORA_DISK_1: restore complete, elapsed time: 00:00:07

Finished restore at 18-MAY-25

RMAN> **recover database;**

Starting recover at 18-MAY-25

using channel ORA_DISK_1

starting media recovery


```

archived log for thread 1 with sequence 25 is already on disk as file
/home/oracle/arch1/arch_1_25_1201218512.arc
archived log for thread 1 with sequence 26 is already on disk as file
/home/oracle/arch1/arch_1_26_1201218512.arc
archived log for thread 1 with sequence 27 is already on disk as file
/home/oracle/arch1/arch_1_27_1201218512.arc
archived log for thread 1 with sequence 28 is already on disk as file
/home/oracle/arch1/arch_1_28_1201218512.arc
archived log for thread 1 with sequence 29 is already on disk as file
/home/oracle/arch1/arch_1_29_1201218512.arc
archived log file name=/home/oracle/arch1/arch_1_25_1201218512.arc thread=1 sequence=25
archived log file name=/home/oracle/arch1/arch_1_26_1201218512.arc thread=1 sequence=26
archived log file name=/home/oracle/arch1/arch_1_27_1201218512.arc thread=1 sequence=27
media recovery complete, elapsed time: 00:00:01
Finished recover at 18-MAY-25

```

RMAN> **alter database open;**

Statement processed

RMAN> **list failure;**

Database Role: PRIMARY

no failures found that match specification

RMAN> **report schema;**

Report of database schema for database with db_unique_name ORCL

List of Permanent Datafiles

=====

File	Size(MB)	Tablespace	RB segs	Datafile Name
1	910	SYSTEM	YES	/u01/app/oracle/oradata/ORCL/system01.dbf
3	600	SYSAUX	NO	/u01/app/oracle/oradata/ORCL/sysaux01.dbf
4	345	UNDOTBS1	YES	/u01/app/oracle/oradata/ORCL/undotbs01.dbf
5	10	SHTBS	NO	/u01/app/oracle/oradata/ORCL/shtbs01.dbf
7	5	USERS	NO	/u01/app/oracle/oradata/ORCL/users01.dbf

List of Temporary Files

=====

File Size(MB) Tablespace Maxsize(MB) Tempfile Name

1	32	TEMP	32767	/u01/app/oracle/oradata/ORCL/temp01.dbf
---	----	------	-------	---

<<시나리오 6>> 모든 DFs의 손상으로 새로운 위치로 복구

```
RMAN> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;
```

```
RMAN> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

```
RMAN> select sequence#, name, FIRST_CHANGE#, NEXT_CHANGE# from v$archived_log;
```

■ 복구 시나리오

```
RMAN> host 'ls /u01/app/oracle/oradata/ORCL/*.dbf';
```

```
/u01/app/oracle/oradata/ORCL/shtbs01.dbf
```

```
/u01/app/oracle/oradata/ORCL/sysaux01.dbf
```

```
/u01/app/oracle/oradata/ORCL/system01.dbf
```

```
/u01/app/oracle/oradata/ORCL/temp01.dbf
```

```
/u01/app/oracle/oradata/ORCL/undotbs01.dbf
```

```
/u01/app/oracle/oradata/ORCL/users01.dbf
```

```
host command complete
```

```
RMAN> host 'rm /u01/app/oracle/oradata/ORCL/*.dbf';
```

```
host command complete
```

```
RMAN> host 'ls /u01/app/oracle/oradata/ORCL/*.dbf';
```

```
ls: cannot access /u01/app/oracle/oradata/ORCL/*.dbf: No such file or directory
```

```
host command complete
```

```
RMAN> alter system checkpoint;
```

```
Statement processed
```

- Alert log file

```
2025-05-18T20:52:08.481333+09:00
```

```
Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl_mz00_3212.trc:
```

```
ORA-01110: data file 1: '/u01/app/oracle/oradata/ORCL/system01.dbf'
```

```
ORA-01565: error in identifying file '/u01/app/oracle/oradata/ORCL/system01.dbf'
```

```
ORA-27037: unable to obtain file status
```

```
Linux-x86_64 Error: 2: No such file or directory
```

```
-----
```

```
RMAN> exit
```

```
[oracle@oel7v9r1 ~]$ rman target "'sys/oracle@t1 as sysdba'" catalog ruser/ruser@c1
```

RMAN> **shutdown abort**

Oracle instance shut down

RMAN> **startup**

connected to target database (not started)

Oracle instance started

database mounted

RMAN> **list failure;**

Database Role: PRIMARY

List of Database Failures

=====

Failure ID	Priority	Status	Time Detected	Summary
6382	CRITICAL	OPEN	18-MAY-25	System datafile 1: '/u01/app/oracle/oradata/ORCL/system01.dbf' is missing
2	HIGH	OPEN	18-MAY-25	One or more non-system datafiles are missing

RMAN> **list failure 6382 detail;**

Database Role: PRIMARY

List of Database Failures

=====

Failure ID	Priority	Status	Time Detected	Summary
6382	CRITICAL	OPEN	18-MAY-25	System datafile 1: '/u01/app/oracle/oradata/ORCL/system01.dbf' is missing

Impact: Database cannot be opened

RMAN> **list failure 2 detail;**

Database Role: PRIMARY

List of Database Failures

=====

Failure ID	Priority	Status	Time Detected	Summary
2	HIGH	OPEN	18-MAY-25	One or more non-system datafiles are missing

Impact: See impact for individual child failures

List of child failures for parent failure ID 2

Failure ID	Priority	Status	Time Detected	Summary
6406	HIGH	OPEN	18-MAY-25	Datafile 7: '/u01/app/oracle/oradata/ORCL/users01.dbf' is missing

Impact: Some objects in tablespace USERS might be unavailable

6400 HIGH OPEN 18-MAY-25 Datafile 5: '/u01/app/oracle/oradata/ORCL/shtbs01.dbf' is missing

Impact: Some objects in tablespace SHTBS might be unavailable

6394 HIGH OPEN 18-MAY-25 Datafile 4: '/u01/app/oracle/oradata/ORCL/undotbs01.dbf' is missing

Impact: Some objects in tablespace UNDOTBS1 might be unavailable

6388 HIGH OPEN 18-MAY-25 Datafile 3: '/u01/app/oracle/oradata/ORCL/sysaux01.dbf' is missing

Impact: Some objects in tablespace SYSAUX might be unavailable

RMAN> **report schema;**

Report of database schema for database with db_unique_name ORCL

List of Permanent Datafiles

=====

File	Size(MB)	Tablespace	RB segs	Datafile Name
1	910	SYSTEM	YES	/u01/app/oracle/oradata/ORCL/system01.dbf
3	600	SYSAUX	NO	/u01/app/oracle/oradata/ORCL/sysaux01.dbf
4	345	UNDOTBS1	YES	/u01/app/oracle/oradata/ORCL/undotbs01.dbf
5	10	SHTBS	NO	/u01/app/oracle/oradata/ORCL/shtbs01.dbf
7	5	USERS	NO	/u01/app/oracle/oradata/ORCL/users01.dbf

List of Temporary Files

=====

File	Size(MB)	Tablespace	Maxsize(MB)	Tempfile Name
1	32	TEMP	32767	/u01/app/oracle/oradata/ORCL/temp01.dbf

- Target DB에서 장애 발생

[oracle@oel7v9r1 ~]\$ **rm -rf /home/oracle/oradata/***

[oracle@oel7v9r1 ~]\$ **ls /home/oracle/oradata/**

[oracle@oel7v9r1 ~]\$

- Recovery

RMAN> **run {**

2> set newname for datafile 1 to '/home/oracle/oradata/system01.dbf';

3> set newname for datafile 4 to '/home/oracle/oradata/undotbs01.dbf';

4> set newname for datafile 3 to '/home/oracle/oradata/sysaux01.dbf';

5> set newname for datafile 5 to '/home/oracle/oradata/shtbs01.dbf';

6> set newname for datafile 7 to '/home/oracle/oradata/users01.dbf';

```
7> set newname for tempfile 1 to '/home/oracle/oradata/temp01.dbf';
8> restore database;
9> switch datafile all;
10> switch tempfile all;
11> recover database;
12> alter database open;
13> }
```

executing command: SET NEWNAME
executing command: SET NEWNAME
executing command: SET NEWNAME
executing command: SET NEWNAME
executing command: SET NEWNAME
executing command: SET NEWNAME

Starting restore at 18-MAY-25
allocated channel: ORA_DISK_1
channel ORA_DISK_1: SID=39 device type=DISK

channel ORA_DISK_1: starting datafile backup set restore
channel ORA_DISK_1: specifying datafile(s) to restore from backup set
channel ORA_DISK_1: restoring datafile 00001 to /home/oracle/oradata/system01.dbf
channel ORA_DISK_1: restoring datafile 00003 to /home/oracle/oradata/sysaux01.dbf
channel ORA_DISK_1: restoring datafile 00004 to /home/oracle/oradata/undotbs01.dbf
channel ORA_DISK_1: restoring datafile 00005 to /home/oracle/oradata/shtbs01.dbf
channel ORA_DISK_1: restoring datafile 00007 to /home/oracle/oradata/users01.dbf
channel ORA_DISK_1: reading from backup piece
/home/oracle/backup/FRA/ORCL/backupset/2025_05_18/o1_mf_nnndf_TAG20250518T021707_n2kk4mp1_.bkp
channel ORA_DISK_1: piece
handle=/home/oracle/backup/FRA/ORCL/backupset/2025_05_18/o1_mf_nnndf_TAG20250518T021707_n2kk4mp1_.
bkp tag=TAG20250518T021707
channel ORA_DISK_1: restored backup piece 1
channel ORA_DISK_1: restore complete, elapsed time: 00:00:07
Finished restore at 18-MAY-25

datafile 1 switched to datafile copy
input datafile copy RECID=8 STAMP=1201468213 file name=/home/oracle/oradata/system01.dbf
datafile 3 switched to datafile copy
input datafile copy RECID=9 STAMP=1201468213 file name=/home/oracle/oradata/sysaux01.dbf
datafile 4 switched to datafile copy
input datafile copy RECID=10 STAMP=1201468213 file name=/home/oracle/oradata/undotbs01.dbf

datafile 5 switched to datafile copy

input datafile copy RECID=11 STAMP=1201468213 file name=/home/oracle/oradata/shtbs01.dbf

datafile 7 switched to datafile copy

input datafile copy RECID=12 STAMP=1201468213 file name=/home/oracle/oradata/users01.dbf

starting full resync of recovery catalog

full resync complete

renamed tempfile 1 to /home/oracle/oradata/temp01.dbf in control file

starting full resync of recovery catalog

full resync complete

Starting recover at 18-MAY-25

using channel ORA_DISK_1

starting media recovery

archived log for thread 1 with sequence 25 is already on disk as file
/home/oracle/arch1/arch_1_25_1201218512.arc

archived log for thread 1 with sequence 26 is already on disk as file
/home/oracle/arch1/arch_1_26_1201218512.arc

archived log for thread 1 with sequence 27 is already on disk as file
/home/oracle/arch1/arch_1_27_1201218512.arc

archived log for thread 1 with sequence 28 is already on disk as file
/home/oracle/arch1/arch_1_28_1201218512.arc

archived log for thread 1 with sequence 29 is already on disk as file
/home/oracle/arch1/arch_1_29_1201218512.arc

archived log for thread 1 with sequence 30 is already on disk as file
/home/oracle/arch1/arch_1_30_1201218512.arc

archived log for thread 1 with sequence 31 is already on disk as file
/home/oracle/arch1/arch_1_31_1201218512.arc

archived log for thread 1 with sequence 32 is already on disk as file
/home/oracle/arch1/arch_1_32_1201218512.arc

archived log for thread 1 with sequence 33 is already on disk as file
/home/oracle/arch1/arch_1_33_1201218512.arc

archived log for thread 1 with sequence 34 is already on disk as file
/home/oracle/arch1/arch_1_34_1201218512.arc

archived log file name=/home/oracle/arch1/arch_1_25_1201218512.arc thread=1 sequence=25

archived log file name=/home/oracle/arch1/arch_1_26_1201218512.arc thread=1 sequence=26

archived log file name=/home/oracle/arch1/arch_1_27_1201218512.arc thread=1 sequence=27

archived log file name=/home/oracle/arch1/arch_1_28_1201218512.arc thread=1 sequence=28

archived log file name=/home/oracle/arch1/arch_1_29_1201218512.arc thread=1 sequence=29

archived log file name=/home/oracle/arch1/arch_1_30_1201218512.arc thread=1 sequence=30
archived log file name=/home/oracle/arch1/arch_1_31_1201218512.arc thread=1 sequence=31
archived log file name=/home/oracle/arch1/arch_1_32_1201218512.arc thread=1 sequence=32
media recovery complete, elapsed time: 00:00:03
Finished recover at 18-MAY-25

Statement processed

SYS@orcl> select status from v\$instance;

STATUS

OPEN

SYS@orcl> select file#, name from v\$datafile;

FILE#	NAME

1	/home/oracle/oradata/system01.dbf
3	/home/oracle/oradata/sysaux01.dbf
4	/home/oracle/oradata/undotbs01.dbf
5	/home/oracle/oradata/shtbs01.dbf
7	/home/oracle/oradata/users01.dbf

RMAN> report obsolete;

RMAN> delete obsolete;

<< 과제 >>

모든DFs의 손상으로 원래 위치로 복구하기

home/oracle/oradata --> /u01/app/oracle/oradata/ORCL/

<<시나리오 7>> 백업받지 않은 TS의 DF 손상 복구

SYS@orcl> create tablespace oetbs datafile '/u01/app/oracle/oradata/ORCL/oetbs01.dbf' size 10m;
 Tablespace created.

SYS@orcl> create table hr.dept1 tablespace oetbs as select * from hr.departments;
 Table created.

SYS@orcl> select count(*) from hr.dept1;
 COUNT(*)

27

RMAN> list backup of datafile "/u01/app/oracle/oradata/ORCL/oetbs01.dbf";
 starting full resync of recovery catalog
 full resync complete
 specification does not match any backup in the repository

RMAN> list backup of tablespace oetbs;
 specification does not match any backup in the repository

RMAN> alter system switch logfile;
 Statement processed

RMAN> alter system switch logfile;
 Statement processed

RMAN> alter system switch logfile;
 Statement processed

- Target DB에서

SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/oetbs01.dbf

SYS@orcl> alter system checkpoint;
 System altered.

SYS@orcl> conn hr/hr
 Connected.

HR@orcl> select count(*) from dept1;
 COUNT(*)

27 <---- 오잉???

```
HR@orcl> !ls /u01/app/oracle/oradata/ORCL
control01.ctl redo01.log shtbs01.dbf temp01.dbf
control02.ctl redo02.log sysaux01.dbf undotbs01.dbf
initorcl.ora redo03.log system01.dbf users01.dbf
```

```
HR@orcl> insert into dept1 select * from dept1;
insert into dept1 select * from dept1
*
```

ERROR at line 1:

```
ORA-01116: error in opening database file 2
ORA-01110: data file 2: '/u01/app/oracle/oradata/ORCL/oetbs01.dbf'
ORA-27041: unable to open file
Linux-x86_64 Error: 2: No such file or directory
Additional information: 3
```

```
RMAN> list failure;
```

Database Role: PRIMARY

List of Database Failures

```
=====
```

Failure ID	Priority	Status	Time Detected	Summary
------------	----------	--------	---------------	---------

```
-----
```

2	HIGH	OPEN	18-MAY-25	One or more non-system datafiles are missing
---	------	------	-----------	--

```
RMAN> list failure 2 detail;
```

Database Role: PRIMARY

List of Database Failures

```
=====
```

Failure ID	Priority	Status	Time Detected	Summary
------------	----------	--------	---------------	---------

```
-----
```

2	HIGH	OPEN	18-MAY-25	One or more non-system datafiles are missing
---	------	------	-----------	--

Impact: See impact for individual child failures

List of child failures for parent failure ID 2

Failure ID	Priority	Status	Time Detected	Summary
------------	----------	--------	---------------	---------

```
-----
```

6605	HIGH	OPEN	18-MAY-25	Datafile 2: '/u01/app/oracle/oradata/ORCL/oetbs01.dbf' is missing
------	------	------	-----------	---

Impact: Some objects in tablespace OETBS might be unavailable

- Recovery

```
RMAN> alter database datafile 2 offline;
```

Statement processed

```
RMAN> alter database create datafile '/u01/app/oracle/oradata/ORCL/oetbs01.dbf';
```

Statement processed

```
RMAN> host 'ls /u01/app/oracle/oradata/ORCL/oetbs01.dbf' ;
```

```
/u01/app/oracle/oradata/ORCL/oetbs01.dbf
```

host command complete

```
RMAN> recover datafile 2;
```

Starting recover at 18-MAY-25

allocated channel: ORA_DISK_1

channel ORA_DISK_1: SID=63 device type=DISK

starting media recovery

archived log for thread 1 with sequence 37 is already on disk as file
/home/oracle/arch1/arch_1_37_1201218512.arc

archived log for thread 1 with sequence 38 is already on disk as file
/home/oracle/arch1/arch_1_38_1201218512.arc

archived log for thread 1 with sequence 39 is already on disk as file
/home/oracle/arch1/arch_1_39_1201218512.arc

archived log file name=/home/oracle/arch1/arch_1_37_1201218512.arc thread=1 sequence=37

media recovery complete, elapsed time: 00:00:00

Finished recover at 18-MAY-25

```
RMAN> alter database datafile 2 online;
```

Statement processed

```
RMAN> report schema;
```

Report of database schema for database with db_unique_name ORCL

List of Permanent Datafiles

=====

File	Size(MB)	Tablespace	RB segs	Datafile Name
1	910	SYSTEM	YES	/u01/app/oracle/oradata/ORCL/system01.dbf
2	10	OETBS	NO	/u01/app/oracle/oradata/ORCL/oetbs01.dbf
3	600	SYSAUX	NO	/u01/app/oracle/oradata/ORCL/sysaux01.dbf
4	345	UNDOTBS1	YES	/u01/app/oracle/oradata/ORCL/undotbs01.dbf
5	10	SHTBS	NO	/u01/app/oracle/oradata/ORCL/shtbs01.dbf

```
7      5      USERS              NO      /u01/app/oracle/oradata/ORCL/users01.dbf
```

List of Temporary Files

=====

File	Size(MB)	Tablespace	Maxsize(MB)	Tempfile Name
1	32	TEMP	32767	/u01/app/oracle/oradata/ORCL/temp01.dbf

```
RMAN> select count(*) from hr.dept1;
```

```

COUNT(*)
-----
        27

```

```
RMAN> select f.tablespace_name, f.file_name from dba_extents e, dba_data_files f where e.file_id = f.file_id and
e.segment_name='DEPT1' and e.owner='HR';
```

TABLESPACE_NAME	FILE_NAME
OETBS	/u01/app/oracle/oradata/ORCL/oetbs01.dbf

```
RMAN> report need backup;
```

RMAN retention policy will be applied to the command

RMAN retention policy is set to redundancy 1

Report of files with less than 1 redundant backups

File #bkps Name

```
-----
2      0      /u01/app/oracle/oradata/ORCL/oetbs01.dbf
```

```
RMAN> backup datafile 2;
```

Starting backup at 18-MAY-25

using channel ORA_DISK_1

channel ORA_DISK_1: starting full datafile backup set

channel ORA_DISK_1: specifying datafile(s) in backup set

input datafile file number=00002 name=/u01/app/oracle/oradata/ORCL/oetbs01.dbf

channel ORA_DISK_1: starting piece 1 at 18-MAY-25

channel ORA_DISK_1: finished piece 1 at 18-MAY-25

piece

handle=/home/oracle/backup/FRA/ORCL/backupset/2025_05_18/o1_mf_nnndf_TAG20250518T234220_n2mwgd7l_

bkp tag=TAG20250518T234220 comment=NONE

channel ORA_DISK_1: backup set complete, elapsed time: 00:00:01

Finished backup at 18-MAY-25

Starting Control File and SPFILE Autobackup at 18-MAY-25

piece handle=/home/oracle/backup/FRA/ORCL/autobackup/2025_05_18/o1_mf_s_1201477342_n2mwggnh_.bkp

comment=NONE

Finished Control File and SPFILE Autobackup at 18-MAY-25

RMAN> **list backup of datafile 2;**

List of Backup Sets

=====

BS Key	Type	LV	Size	Device Type	Elapsed Time	Completion Time
--------	------	----	------	-------------	--------------	-----------------

1431	Full		1.09M	DISK	00:00:00	18-MAY-25
------	------	--	-------	------	----------	-----------

BP Key: 1434 Status: AVAILABLE Compressed: NO Tag: TAG20250518T234220

Piece

Name:

/home/oracle/backup/FRA/ORCL/backupset/2025_05_18/o1_mf_nnndf_TAG20250518T234220_n2mwgd7l_.bkp

List of Datafiles in backup set 1431

File	LV	Type	Ckp	SCN	Ckp Time	Abs Fuz	SCN	Sparse	Name
------	----	------	-----	-----	----------	---------	-----	--------	------

2		Full	4381331		18-MAY-25			NO	/u01/app/oracle/oradata/ORCL/oetbs01.dbf
---	--	------	---------	--	-----------	--	--	----	--

<<시나리오 8>> TS에 속한 여러 DF중 손상된 경우 복구

```

RMAN> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;

```

FILE#	TBS_NAME	FILE_NAME	CHECKPOINT_CHANGE#	STATUS
3	SYSAUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	4379323	ONLINE
1	SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf	4379323	SYSTEM
4	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	4379323	ONLINE
7	USERS	/u01/app/oracle/oradata/ORCL/users01.dbf	4379323	ONLINE
5	SHTBS	/u01/app/oracle/oradata/ORCL/shtbs01.dbf	4379323	ONLINE
2	OETBS	/u01/app/oracle/oradata/ORCL/oetbs01.dbf	4381331	ONLINE

```

RMAN> alter tablespace oetbs add datafile '/u01/app/oracle/oradata/ORCL/oetbs02.dbf' size 10m;
Statement processed
starting full resync of recovery catalog
full resync complete

```

```

RMAN> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;

```

FILE#	TBS_NAME	FILE_NAME	CHECKPOINT_CHANGE#	STATUS
3	SYSAUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	4379323	ONLINE
1	SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf	4379323	SYSTEM
4	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	4379323	ONLINE
7	USERS	/u01/app/oracle/oradata/ORCL/users01.dbf	4379323	ONLINE
5	SHTBS	/u01/app/oracle/oradata/ORCL/shtbs01.dbf	4379323	ONLINE
2	OETBS	/u01/app/oracle/oradata/ORCL/oetbs01.dbf	4381331	ONLINE
8	OETBS	/u01/app/oracle/oradata/ORCL/oetbs02.dbf	4381580	ONLINE

7 rows selected

```

SYS@orcl> create table hr.emp3 tablespace oetbs as select * from hr.employees;
Table created.

```

```

SYS@orcl> select f.tablespace_name, f.file_name from dba_extents e, dba_data_files f where e.file_id = f.file_id and
e.segment_name='EMP3' and e.owner='HR';

```

TABLESPACE_NAME	FILE_NAME
OETBS	/u01/app/oracle/oradata/ORCL/oetbs01.dbf

```
SYS@orcl> select f.tablespace_name, f.file_name from dba_extents e, dba_data_files f where e.file_id = f.file_id and e.segment_name='DEPT1' and e.owner='HR';
```

TABSPACE_NAME	FILE_NAME

OETBS	/u01/app/oracle/oradata/ORCL/oetbs01.dbf

```
SYS@orcl> insert into hr.emp3 select * from hr.emp3;  
107 rows created.
```

```
SYS@orcl> /  
214 rows created.
```

```
SYS@orcl> /  
428 rows created.
```

```
SYS@orcl> /  
856 rows created.
```

```
SYS@orcl> /  
1712 rows created.
```

```
SYS@orcl> /  
3424 rows created.
```

```
SYS@orcl> /  
6848 rows created.
```

```
SYS@orcl> /  
13696 rows created.
```

```
SYS@orcl> /  
27392 rows created.
```

```
SYS@orcl> select f.tablespace_name, f.file_name from dba_extents e, dba_data_files f where e.file_id = f.file_id and e.segment_name='EMP3' and e.owner='HR';
```

TABSPACE_NAME	FILE_NAME

OETBS	/u01/app/oracle/oradata/ORCL/oetbs01.dbf
OETBS	/u01/app/oracle/oradata/ORCL/oetbs02.dbf

RMAN> report schema;

RMAN> report need backup;

RMAN retention policy will be applied to the command

RMAN retention policy is set to redundancy 1

Report of files with less than 1 redundant backups

File #bkps Name

```
-----
8      0      /u01/app/oracle/oradata/ORCL/oetbs02.dbf
```

RMAN> backup datafile 8;

Starting backup at 19-MAY-25

using channel ORA_DISK_1

channel ORA_DISK_1: starting full datafile backup set

channel ORA_DISK_1: specifying datafile(s) in backup set

input datafile file number=00008 name=/u01/app/oracle/oradata/ORCL/oetbs02.dbf

channel ORA_DISK_1: starting piece 1 at 19-MAY-25

channel ORA_DISK_1: finished piece 1 at 19-MAY-25

piece

handle=/home/oracle/backup/FRA/ORCL/backupset/2025_05_19/o1_mf_nnndf_TAG20250519T000205_n2mxmfsk_b

kp tag=TAG20250519T000205 comment=NONE

channel ORA_DISK_1: backup set complete, elapsed time: 00:00:01

Finished backup at 19-MAY-25

Starting Control File and SPFILE Autobackup at 19-MAY-25

piece handle=/home/oracle/backup/FRA/ORCL/autobackup/2025_05_19/o1_mf_s_1201478527_n2mxmhqh_.bkp

comment=NONE

Finished Control File and SPFILE Autobackup at 19-MAY-25

RMAN> list backup of datafile 8;

List of Backup Sets

=====

BS Key	Type	LV	Size	Device	Type	Elapsed Time	Completion Time
--------	------	----	------	--------	------	--------------	-----------------

1527	Full		2.65M	DISK		00:00:00	19-MAY-25
------	------	--	-------	------	--	----------	-----------

BP Key: 1530 Status: AVAILABLE Compressed: NO Tag: TAG20250519T000205

Piece

Name:

/home/oracle/backup/FRA/ORCL/backupset/2025_05_19/o1_mf_nnndf_TAG20250519T000205_n2mxmfsk_.bkp

List of Datafiles in backup set 1527

File	LV	Type	Ckp	SCN	Ckp Time	Abs Fuz	SCN	Sparse	Name
------	----	------	-----	-----	----------	---------	-----	--------	------

8		Full	4382490		19-MAY-25			NO	/u01/app/oracle/oradata/ORCL/oetbs02.dbf
---	--	------	---------	--	-----------	--	--	----	--

■ 복구 시나리오

- 장애 발생

```
SYS@orcl> !rm /u01/app/oracle/oradata/ORCL/oetbs02.dbf
```

```
SYS@orcl> alter system checkpoint;
```

System altered.

```
SYS@orcl> conn hr/hr
```

Connected.

```
HR@orcl> select count(*) from dept1;
```

COUNT(*)

27

```
HR@orcl> select count(*) from emp3;
```

COUNT(*)

54784

```
HR@orcl> conn / as sysdba
```

Connected.

```
SYS@orcl> !ls /u01/app/oracle/oradata/ORCL/
```

```
control01.ctl  oetbs01.dbf  redo03.log      system01.dbf  users01.dbf
```

```
control02.ctl  redo01.log   shtbs01.dbf    temp01.dbf
```

```
initorcl.ora   redo02.log   sysaux01.dbf   undotbs01.dbf
```

```
SYS@orcl> shut abort
```

ORACLE instance shut down.

```
SYS@orcl> startup
```

ORACLE instance started.

Total System Global Area 1140849904 bytes

Fixed Size 8895728 bytes

Variable Size 822083584 bytes

Database Buffers 301989888 bytes

Redo Buffers 7880704 bytes

Database mounted.

ORA-01157: cannot identify/lock data file 8 - see DBWR trace file

ORA-01110: data file 8: '/u01/app/oracle/oradata/ORCL/oetbs02.dbf'

- Alert log file

2025-05-19T00:07:55.926484+09:00

Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl_mz00_10321.trc:

ORA-01110: data file 8: '/u01/app/oracle/oradata/ORCL/oetbs02.dbf'
 ORA-01565: error in identifying file '/u01/app/oracle/oradata/ORCL/oetbs02.dbf'
 ORA-27037: unable to obtain file status
 Linux-x86_64 Error: 2: No such file or directory
 Additional information: 7

RMAN> **exit**

[oracle@oel7v9r1 ~]\$ **rman target "sys/oracle@t1 as sysdba" catalog ruser/ruser@c1**

RMAN> **list failure;**

Database Role: PRIMARY

List of Database Failures

=====

Failure ID	Priority	Status	Time Detected	Summary
------------	----------	--------	---------------	---------

2	HIGH	OPEN	19-MAY-25	One or more non-system datafiles are missing
---	------	------	-----------	--

RMAN> **list failure 2 detail;**

Database Role: PRIMARY

List of Database Failures

=====

Failure ID	Priority	Status	Time Detected	Summary
------------	----------	--------	---------------	---------

2	HIGH	OPEN	19-MAY-25	One or more non-system datafiles are missing
---	------	------	-----------	--

Impact: See impact for individual child failures

List of child failures for parent failure ID 2

Failure ID	Priority	Status	Time Detected	Summary
------------	----------	--------	---------------	---------

6745	HIGH	OPEN	19-MAY-25	Datafile 8: '/u01/app/oracle/oradata/ORCL/oetbs02.dbf' is missing
------	------	------	-----------	---

Impact: Some objects in tablespace OETBS might be unavailable

RMAN> **report schema;**

RMAN> **alter database datafile 8 offline;**

Statement processed

RMAN> **alter database open;**

Statement processed

RMAN> **select count(*) from hr.dept1;**

COUNT(*)

27

RMAN> **select count(*) from hr.emp3;**

RMAN-00571: =====

RMAN-00569: ===== ERROR MESSAGE STACK FOLLOWS =====

RMAN-00571: =====

RMAN-03002: failure of sql statement command at 05/19/2025 00:13:55

ORA-00376: file 8 cannot be read at this time

ORA-01110: data file 8: '/u01/app/oracle/oradata/ORCL/oetbs02.dbf'

-Recovery

RMAN> **restore datafile 8;**

Starting restore at 19-MAY-25

allocated channel: ORA_DISK_1

channel ORA_DISK_1: SID=70 device type=DISK

channel ORA_DISK_1: starting datafile backup set restore

channel ORA_DISK_1: specifying datafile(s) to restore from backup set

channel ORA_DISK_1: restoring datafile 00008 to /u01/app/oracle/oradata/ORCL/oetbs02.dbf

channel	ORA_DISK_1:	reading	from	backup	piece
/home/oracle/backup/FRA/ORCL/backupset/2025_05_19/o1_mf_nnndf_TAG20250519T000205_n2mxmfsk_bkp					

channel	ORA_DISK_1:				piece
---------	-------------	--	--	--	-------

handle=/home/oracle/backup/FRA/ORCL/backupset/2025_05_19/o1_mf_nnndf_TAG20250519T000205_n2mxmfsk_bkp tag=TAG20250519T000205

channel ORA_DISK_1: restored backup piece 1

channel ORA_DISK_1: restore complete, elapsed time: 00:00:01

Finished restore at 19-MAY-25

RMAN> **recover datafile 8;**

Starting recover at 19-MAY-25

using channel ORA_DISK_1

starting media recovery

media recovery complete, elapsed time: 00:00:00

Finished recover at 19-MAY-25

RMAN> **alter database datafile 8 online;**

Statement processed

RMAN> **report schema;**

Report of database schema for database with db_unique_name ORCL

List of Permanent Datafiles

=====

File	Size(MB)	Tablespace	RB segs	Datafile Name
1	910	SYSTEM	YES	/u01/app/oracle/oradata/ORCL/system01.dbf
2	10	OETBS	NO	/u01/app/oracle/oradata/ORCL/oetbs01.dbf
3	610	SYSAUX	NO	/u01/app/oracle/oradata/ORCL/sysaux01.dbf
4	345	UNDOTBS1	YES	/u01/app/oracle/oradata/ORCL/undotbs01.dbf
5	10	SHTBS	NO	/u01/app/oracle/oradata/ORCL/shtbs01.dbf
7	5	USERS	NO	/u01/app/oracle/oradata/ORCL/users01.dbf
8	10	OETBS	NO	/u01/app/oracle/oradata/ORCL/oetbs02.dbf

List of Temporary Files

=====

File	Size(MB)	Tablespace	Maxsize(MB)	Tempfile Name
1	32	TEMP	32767	/u01/app/oracle/oradata/ORCL/temp01.dbf

RMAN> **select count(*) from hr.emp3;**

COUNT(*)

54784